Journal of the Royal Statistical Society Series C: Applied Statistics, , , 1–36

Supplementary theory and analyses for "Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China"

Jian Hou,[1] **Tan Meng**[2] **and Maozai Tian**[2,*]

[1]College of Systems Engineering, National University of Defense Technology, Changsha 410073, China

[2]Center for Applied Statistics, School of Statistics, Renmin University of China, 59 Zhongguancun Street, Beijing 100872, China

[*]Corresponding author. mztian@ruc.edu.cn

This supplement provides the proof of Proposition 1, a process-level strong-mixing result, additional applied analyses and the simulation specifications and results. Unless stated otherwise, evaluation-period tables use 33 summers and exclude the 2015 and 2022 method-development summers. The historical and station extensions state their own temporal supports.

Analysis chronology

Table S1 summarises the sequence of analyses. Unless stated otherwise, each calculation uses the 121-site labels, five bandwidths and equal-month, equal-scale and equal-summer estimator. The historical and station extensions were designed before their respective data were retrieved. Table S1 distinguishes the primary analyses from subsequent comparisons and states the assumptions used for inference.

The original Version 2 protocol, retained as `protocols/archive/CONFIRMATORY_ANALYSIS_PROTOCOL.md` in the reproducibility archive, states a freeze date of 2 August 2026, before acquisition or inspection of the 33 evaluation summers. Its SHA-256 is `13c3a32532921e9230d00e88e07bb1aaa95e1a27ac970d843739bbb3090a3a70`. The first commit visible in the public Git history is later, on 11 August 2026 (2464330). The protocol date is recorded in the file; the public record begins with that later commit. The submission release at `https://github.com/Stork343/eastern-china-wet-bulb-contraction/releases/tag/jrssc-submission-v1-2026-08-13` archives the protocol, its digest, this status table and the submitted analysis together.

We developed the method using the complete 2015 and 2022 summers, inspecting their wet-bulb fields and graph profiles. The available records leave the original acquisition rationale undocumented. Evaluation uses the other 33 summers.

Table S1. Protocol status, timing, observed result and final interpretive role.

Element	Original status	Fixed before the 33-summer results?	Observed result and final role	Reason or scope
Field construction, event labels, five graph scales and aggregation	Primary specification	Yes	Retained; defines the finite-record target	Developed using 2015 and 2022, which were excluded from evaluation
33-summer evaluation-period contrast	Primary estimand	Yes	-7.2834%; primary finite-record summary	Equal months, scales and summers
Ordinary Student mean test	Confirmatory component	Yes	One-sided $p = 2.56 \times 10^{-6}$; process reference	Later simulations showed undercoverage under appreciable interannual dependence
Lag-2 Newey–West test	Confirmatory component	Yes	One-sided $p = 3.54 \times 10^{-6}$; process reference	Consistency depends on the omitted autocovariances
Independent-summer exact binomial sign reference	Confirmatory component	Yes	28/33 negative; $p = 3.31 \times 10^{-5}$; recurrence reference	Its exact-binomial interpretation requires independent summer signs
Three-component consistency rule	Confirmatory rule	Yes	All three p-values below 0.025	Student and HAC coverage depends on interannual correlation
Global 99,999-draw product shift	Not in the protocol	No; added after observing the primary result	Exploratory conditional randomisation; $p_{\mathrm{MC}} = 10^{-5}$	Conditional product-invariance null
Process-model references, alternative effect measures, spatial bases, bandwidth and support comparisons	Exploratory analyses	No, except where separately dated	Reference inference, structural descriptions and sensitivity analyses	Comparisons with the primary raw-ratio summary
1950–1990 ERA5-Land extension	Pre-access extension	Before access to historical values	Retained as a temporally separated extension	Shares the ERA5-Land system with the evaluation record
Historical climatology–anomaly decomposition	Exploratory structural analysis	No	Period-specific algebraic accounting	Climatological alignment and anomaly energy
Ten-year NOAA–ERA5-Land extension	Pre-access extension	Before station results were retrieved	Sparse external measurement comparison	Matched events defined using ERA5-Land
Heavy-tail, coupled-field and cross-record simulations	Exploratory simulations	After method development	Estimator behaviour and conditional-randomisation calibration	Test size under the specified dependence models

S1. Conditional validity of the randomisation test

Let X collect the regional-mean records used to define the event labels, and let D collect the corresponding daily graph-dispersion profiles. The finite group G acts on D by independently rotating the complete multiscale profile within each month-year record. Conditional on X, suppose

$$(D \mid X) \overset{d}{=} (gD \mid X) \quad \text{for every } g \in G. \tag{S1}$$

Conditional on the observed regional-mean series, independently changing the circular phase of the complete dispersion profile in any month-year record must leave the joint distribution unchanged. This conditional phase-invariance assumption specifies the distribution of the full profiles, beyond the mean restriction $\mathrm{E}(Q^E - Q^M) = 0$. Common weather modes across months, circulation or soil-moisture persistence, and other shared seasonal or interannual drivers can violate the product condition because independent record-wise rotations break their phase alignment.

For a measurable lower-tail statistic $T(X, D)$ fixed before the transformations are examined, define

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \leq T(X, D)\}.$$

Proposition 1 (Product cyclic-shift validity) *For every $u \in [0, 1]$,*

$$\Pr\{p_G(X, D) \leq u \mid X\} \leq u \quad \textit{almost surely.}$$

If the $|G|$ transformed statistics have no ties, p_G is uniform on $\{1/|G|, \ldots, 1\}$ conditional on the invariant orbit. If $H_1, \ldots, H_B$ are independent uniforms on G, independently of (X, D), then

$$p_B = \frac{1 + \sum_{b=1}^{B} I\{T(X, H_b D) \leq T(X, D)\}}{B + 1}$$

satisfies

$$\Pr\{p_B \leq u \mid X\} \leq \frac{\lfloor (B+1)u \rfloor}{B+1} \leq u.$$

For fixed scale statistics $T_1, \ldots, T_H$, put $M(X, D) = \min_h T_h(X, D)$ and define

$$p_h^{\text{adj}} = \frac{1}{|G|} \sum_{g \in G} I\{M(X, gD) \leq T_h(X, D)\}.$$

Under the complete joint invariance null, $\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u$. This result provides weak family-wise control for the complete joint null.

Proof Fix a value x in the probability-one set on which the conditional invariance in Equation (S1) holds, and write $S(y) = T(x, y)$. Let $m = |G|$.

For fixed y, form the multiset $\{S(gy) : g \in G\}$. For each $g \in G$, let

$$r_g(y) = \sum_{h \in G} I\{S(hy) \leq S(gy)\}.$$

Right multiplication by g permutes the elements of G. Hence

$$\begin{aligned} p_G(x, gy) &= \frac{1}{m} \sum_{h \in G} I\{S(hgy) \leq S(gy)\} \\ &= \frac{r_g(y)}{m}. \end{aligned}$$

If U is uniform on G, independently of y, then

$$\Pr\{p_G(x, Uy) \leq u \mid y\} = \frac{1}{m} \#\{g : r_g(y) \leq mu\}. \tag{S2}$$

Order the m values in the multiset. An observation at the upper end of a tie block has weak rank equal to the final position of that block. Every tie block counted on the right of Equation (S2) therefore ends no later than position $\lfloor mu \rfloor$. The number of counted group elements is at most $\lfloor mu \rfloor$, and

$$\Pr\{p_G(x, Uy) \leq u \mid y\} \leq \frac{\lfloor mu \rfloor}{m} \leq u.$$

This inequality also holds when the action has a nontrivial stabiliser. Repeated transformations then appear as ties in the multiset and make the weak-rank test conservative.

Let U now be independent of (X, D) and uniform on G. Conditional invariance gives $UD \mid X = x \overset{d}{=} D \mid X = x$. Applying the preceding inequality conditionally on D yields

$$\begin{aligned}\Pr\{p_G(x, D) \le u \mid X = x\} &= \Pr\{p_G(x, UD) \le u \mid X = x\}\\ &= \mathrm{E}[\Pr\{p_G(x, UD) \le u \mid X = x, D\} \mid X = x]\\ &\le u.\end{aligned}$$

If the transformed statistics have no ties, the ranks $r_g(y)$ are $1, \dots, m$ once each. Equality holds at $u = j/m$, proving uniformity on the attainable grid.

For the sampled version, take $U_0, U_1, \dots, U_B$ independently and uniformly from G, independently of (X, D). Conditional on (X, D), the variables $S(U_0D), \dots, S(U_BD)$ are identically distributed and exchangeable. The weak rank of the first among these $B + 1$ values satisfies

$$\Pr\left[\frac{1 + \sum_{b=1}^{B} I\{S(U_bD) \le S(U_0D)\}}{B+1} \le u \;\middle|\; X = x\right] \le \frac{\lfloor (B+1)u \rfloor}{B+1}. \tag{S3}$$

It remains to relate the random anchor to the implemented construction, whose first transformation is the identity. Put $H_b = U_bU_0^{-1}$ for $b = 1, \dots, B$ and $D^* = U_0D$. The map

$$(U_0, U_1, \dots, U_B) \longmapsto (U_0, H_1, \dots, H_B)$$

is a bijection on G^{B+1}. Thus $H_1, \dots, H_B$ are independent uniforms and are independent of U_0. Moreover, $D^* \mid X = x \overset{d}{=} D \mid X = x$. The vector in Equation (S3) consequently has the same conditional distribution as

$$\{S(D), S(H_1D), \dots, S(H_BD)\}.$$

This is the identity-plus-B construction used to calculate p_B.

The statistic must remain a fixed measurable function of the jointly shifted outcome matrix. For the scale-wise single-step comparison, let T_h denote the scale statistics and put $M(y) = \min_h T_h(x, y)$. The adjusted value

$$p_h^{\mathrm{adj}} = \frac{1}{m} \sum_{g \in G} I\{M(gy) \le T_h(x, y)\}$$

is nondecreasing in its observed threshold. Therefore $\min_h p_h^{\mathrm{adj}}$ is attained at a scale for which $T_h(x, y) = M(y)$ and equals the randomisation value for M. Applying the enumeration result to M gives

$$\Pr\{\min_h p_h^{\mathrm{adj}} \le u \mid X\} \le u.$$

The conclusion uses the complete joint null because M combines every scale. This completes the proof. □

S2. Asymptotic inference under strong mixing

Let $\widehat{\boldsymbol{R}}_r = (\widehat{R}_{h_1 r}, \dots, \widehat{R}_{h_H r})^\top$ be the scale profile for summer r, $\boldsymbol{a} = H^{-1}\mathbf{1}_H$, $Z_r = \boldsymbol{a}^\top \widehat{\boldsymbol{R}}_r$ and $\widehat{\theta}_R = R^{-1}\sum_{r=1}^{R} Z_r$. Write $\gamma(k) = \mathrm{Cov}(Z_0, Z_k)$ and define

$$\widehat{\gamma}_R(k) = \frac{1}{R}\sum_{r=k+1}^{R} (Z_r - \overline{Z}_R)(Z_{r-k} - \overline{Z}_R),$$

$$\widehat{\sigma}_R^2 = \widehat{\gamma}_R(0) + 2\sum_{k=1}^{L_R} K\left(\frac{k}{L_R + 1}\right)\widehat{\gamma}_R(k). \tag{S4}$$

Theorem S1 (Strong-mixing limit and HAC reference) *Suppose $\{\widehat{\boldsymbol{R}}_r\}_{r\in\mathbb{Z}}$ is strictly stationary and strongly mixing, with $\mathrm{E}\|\widehat{\boldsymbol{R}}_0\|^{2+\delta} < \infty$ for some $\delta > 0$ and $\sum_{k=1}^{\infty} \alpha(k)^{\delta/(2+\delta)} < \infty$. Then every entry of*

$$\boldsymbol{\Omega} = \sum_{k=-\infty}^{\infty} \mathrm{Cov}(\widehat{\boldsymbol{R}}_0, \widehat{\boldsymbol{R}}_k)$$

is absolutely convergent. If $\sigma^2 = \boldsymbol{a}^\top \boldsymbol{\Omega} \boldsymbol{a} > 0$ and $\theta = \boldsymbol{a}^\top \mathrm{E}(\widehat{\boldsymbol{R}}_0)$, then

$$\sqrt{R}(\widehat{\theta}_R - \theta) \xrightarrow{d} N(0, \sigma^2).$$

Suppose in addition that K is bounded on $[0,\infty)$, $K(0) = 1$, is continuous at zero and vanishes outside $[0,1]$; $L_R \to \infty$, $L_R/R \to 0$; and

$$\sum_{k=0}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1). \tag{S5}$$

Then $\widehat{\sigma}_R^2 \xrightarrow{p} \sigma^2$ and

$$\frac{\sqrt{R}(\widehat{\theta}_R - \theta)}{\widehat{\sigma}_R} \xrightarrow{d} N(0,1).$$

For a ratio component $\widehat{R}_{jr} = A_{jr}/B_{jr} - 1$, with $A_{jr} \geq 0$ and $B_{jr} > 0$, a sufficient condition for its $(2+\delta)$ moment is the existence of conjugate $p, q > 1$ such that

$$\mathrm{E}\, A_{j0}^{(2+\delta)p} < \infty, \qquad \mathrm{E}\, B_{j0}^{-(2+\delta)q} < \infty.$$

The same conclusion holds for a finite average of such ratios when the condition holds for every constituent ratio.

Proof Let $\boldsymbol{\mu} = \mathrm{E}(\widehat{\boldsymbol{R}}_0)$. A measurable transformation cannot increase strong-mixing coefficients, so $\{Z_r\}$ is strictly stationary and strongly mixing with coefficients bounded by $\alpha(k)$. The inequality $|\boldsymbol{a}^\top v| \leq \|\boldsymbol{a}\|\|v\|$ gives $\mathrm{E}\,|Z_0|^{2+\delta} < \infty$.

Put $\xi_r = Z_r - \mathrm{E}\, Z_0$. Davydov's covariance inequality, with both moment exponents equal to $2+\delta$, gives a finite constant C such that

$$|\gamma(k)| = |\,\mathrm{E}(\xi_0 \xi_k)| \leq C\|\xi_0\|_{2+\delta}^2 \alpha(|k|)^{\delta/(2+\delta)}, \qquad k \neq 0. \tag{S6}$$

The mixing-rate assumption implies $\sum_{k=-\infty}^{\infty} |\gamma(k)| < \infty$. Applying the same inequality componentwise establishes absolute convergence of every entry of $\boldsymbol{\Omega}$. Bilinearity and absolute convergence permit interchange of the finite quadratic form and the infinite sum:

$$\boldsymbol{a}^\top \boldsymbol{\Omega} \boldsymbol{a} = \sum_{k=-\infty}^{\infty} \mathrm{Cov}(Z_0, Z_k) = \sum_{k=-\infty}^{\infty} \gamma(k) = \sigma^2. \tag{S7}$$

The central limit theorem for strongly mixing sequences applies under the stated moment and mixing-rate conditions (Ibragimov and Linnik, 1971). Hence

$$R^{-1/2} \sum_{r=1}^{R} \xi_r \xrightarrow{d} N(0, \sigma^2).$$

Since $\mathrm{E}\, Z_0 = \boldsymbol{a}^\top \boldsymbol{\mu} = \theta$ and $R^{-1} \sum_r Z_r = \widehat{\theta}_R$, this proves the first limit.

For consistency of Equation (S4), define

$$\sigma^2_{K,L_R} = \gamma(0) + 2\sum_{k=1}^{L_R} K\left(\frac{k}{L_R+1}\right)\gamma(k).$$

Boundedness of K and Equation (S5) give

$$\begin{aligned}|\widehat{\sigma}^2_R - \sigma^2_{K,L_R}| &\le |\widehat{\gamma}_R(0) - \gamma(0)| \\ &\quad + 2\|K\|_\infty \sum_{k=1}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1).\end{aligned}$$

For each fixed k, continuity at zero and $L_R \to \infty$ imply $K\{k/(L_R+1)\} \to 1$. The weights are uniformly bounded, vanish for $k > L_R$, and $\sum_k |\gamma(k)| < \infty$. Dominated convergence for series therefore yields

$$\sigma^2_{K,L_R} \longrightarrow \gamma(0) + 2\sum_{k=1}^{\infty}\gamma(k) = \sigma^2.$$

Thus $\widehat{\sigma}^2_R \xrightarrow{p} \sigma^2$. Positivity of σ^2 and Slutsky's theorem give the studentised limit.

Condition (S5) is separate from the assumptions used for the central limit theorem. Primitive sufficient conditions for it require higher moments, stronger mixing-rate summability and a compatible bandwidth rate (Andrews, 1991; Newey and West, 1987).

For comparison, fix a lag L. Strong mixing implies ergodicity. The ergodic theorem applied to $(Z_r - \mathrm{E}\, Z_0)(Z_{r-k} - \mathrm{E}\, Z_0)$, together with $\overline{Z}_R \xrightarrow{p} \mathrm{E}\, Z_0$, gives $\widehat{\gamma}_R(k) \xrightarrow{p} \gamma(k)$ for each fixed k. The fixed-lag estimator therefore converges to

$$\sigma^2_{K,L} = \gamma(0) + 2\sum_{k=1}^{L} K\left(\frac{k}{L+1}\right)\gamma(k). \tag{S8}$$

For Bartlett lag $L = 2$, this is $\gamma(0) + 2\{(2/3)\gamma(1) + (1/3)\gamma(2)\}$. Its difference from the full long-run variance is

$$\sigma^2 - \sigma^2_{K,2} = 2\left\{\frac{1}{3}\gamma(1) + \frac{2}{3}\gamma(2) + \sum_{k=3}^{\infty}\gamma(k)\right\}.$$

The fixed-lag estimator is consistent only if this weighted error is zero.

It remains to verify the stated sufficient ratio condition. With $s = 2+\delta$, convexity and Hölder's inequality give

$$\begin{aligned}\mathrm{E}\,|A_{j0}/B_{j0} - 1|^s &\le 2^{s-1}\left\{\mathrm{E}(A^s_{j0}B^{-s}_{j0}) + 1\right\} \\ &\le 2^{s-1}\left[\{\mathrm{E}\, A^{sp}_{j0}\}^{1/p}\{\mathrm{E}\, B^{-sq}_{j0}\}^{1/q} + 1\right] < \infty.\end{aligned}$$

A finite sum of variables with finite sth moments also has a finite sth moment. Applying this argument to each component proves the final claim. □

The 495 observed middle-day denominators range from 2.25 to 25.99 °C^2. The process-level ratio argument additionally assumes the negative moment stated in Theorem S1.

S3. The randomisation distribution

The product-shift analysis used the statistic $T = \widehat{\theta}_{\mathcal{Y}_H}$. Each draw independently selected one cyclic offset in each of the 99 evaluation month-year records and shifted all five graph-dispersion columns together. Regional-mean series and event labels remained in their observed order. The statistic was then reconstructed with equal months, scales and summers. The generator seed was 20260809 and $B = 99{,}999$. This analysis was added after the primary 33-summer result was observed and is reported as exploratory.

No sampled statistic was at or below the observed -7.2834%; the lower-tail exceedance count was zero. The plus-one Monte Carlo value is therefore $p_{\mathrm{MC}} = (0+1)/(99{,}999+1) = 1.0 \times 10^{-5}$, the minimum attainable value with 99,999 draws. The sampled null had mean 0.8241%, standard deviation 1.5651 points and median 0.8139%; its 0.1%, 1% and 5% quantiles were -3.8788%, -2.7440% and -1.7225%. A ratio-based randomisation distribution has its own empirical centre, and the test compares the observed statistic directly with that distribution.

Table S2. The 99,999-draw product cyclic-shift analysis. Effects, null means and null standard deviations are in percentage points. The last column uses the minimum statistic over the five scales and reports weak family-wise validity under the complete joint invariance null.

Bandwidth (km)	Observed (points)	Null mean (points)	Null SD (points)	Unadjusted p	Complete-null p
126	-2.856	0.359	1.059	0.00104	0.02504
252	-3.945	0.531	1.301	0.00022	0.00571
503	-5.907	0.859	1.683	0.00002	0.00019
1,006	-10.441	1.125	1.964	0.00001	0.00001
2,013	-13.268	1.246	2.089	0.00001	0.00001
Five-scale mean	-7.283	0.824	1.565	0.00001	–

The test evaluates the conditional product-invariance null in Equation (S1). Scale-specific rows use the same joint draws and are mutually dependent. Section S11.4 examines test size under a shared process across month-year records.

S4. Temperature interpretation of graph dispersion

For a field $\mathbf{y}$,

$$Q_h(\mathbf{y}) = \frac{\sum_{i<j} w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} w_{ij,h}}, \qquad D_h^{\mathrm{RMS}}(\mathbf{y}) = \{2Q_h(\mathbf{y})\}^{1/2}.$$

Thus Q_h is one half of the edge-weighted mean squared pairwise difference, equivalently an edge-weighted semivariance. It has units °C^2, whereas D_h^{RMS} is the weighted root mean squared pairwise difference in °C. Table S3 reports equal-month and equal-summer means of Q_h and applies $D_h^{\mathrm{RMS}} = (2Q_h)^{1/2}$ to each displayed mean. The relative columns average record-level ratios and therefore differ from ratios of the displayed marginal means.

Table S3. Physical scale of graph dispersion in the evaluation summers. The RMS columns are $(2Q_h)^{1/2}$ transformations of the displayed mean Q_h columns. The interval is a 95% (t)-reference interval under an independent-summer model. RMS change is the mean record-level relative change in D_h^{RMS}.

Bandwidth (km)	Q_h^M (°C^2)	Q_h^E (°C^2)	RMS^M (°C)	RMS^E (°C)	Q_h effect [interval] (%)	RMS change (%)
126	2.662	2.578	2.307	2.271	-2.86 $[-5.04, -0.67]$	-1.54
252	4.356	4.169	2.952	2.887	-3.94 $[-6.50, -1.39]$	-2.14
503	7.943	7.433	3.986	3.856	-5.91 $[-8.91, -2.90]$	-3.25
1,006	12.892	11.489	5.078	4.793	-10.44 $[-13.49, -7.39]$	-5.70
2,013	15.903	13.743	5.640	5.243	-13.27 $[-16.28, -10.26]$	-7.22

The direct five-scale mean of the record-level RMS changes is -3.97%. Applying the square-root transformation only after averaging the five graph-dispersion effects gives $\sqrt{1 - 0.072834} - 1 = -3.71\%$. The two values differ because the square root and the averaging steps do not commute. In either form, a percentage change in squared pairwise differences is larger in magnitude than the corresponding change in RMS amplitude.

The five distances specify the Gaussian bandwidths h. At $h = 126, 252, 503, 1{,}006$ and $2{,}013$ km, the edge-weighted mean pair distances are respectively 200, 319, 549, 826 and 992 km; the corresponding Kish effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. The five-scale summary is an equal-weight mean over this logarithmic bandwidth grid.

S5. Structural decomposition

S5.1. Alignment with the monthly climatology

Let $\boldsymbol{\mu}_m$ be the site-specific 35-summer mean field for month m and write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$. For a fixed graph,

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \tag{S9}$$

The first term is fixed within month m, so its high-minus-middle difference is exactly zero. Within each record, we divide the differences of the other two terms by the middle-day total Q_h mean. The resulting anomaly-energy and climatology–anomaly alignment components add to the original relative contrast before any aggregation.

The middle term measures the signed alignment of daily anomalies with the monthly climatology. Opposing patterns give a negative contribution.

Table S4. Exact climatology–anomaly decomposition. Entries are percentage points; brackets are 95% descriptive (t)-reference intervals based on between-summer variation. The climatology-energy difference is zero by construction.

Bandwidth (km)	Total effect [interval]	Anomaly energy [interval]	Climatology–anomaly alignment [interval]
126	-2.86 $[-5.04, -0.67]$	-2.63 $[-3.55, -1.72]$	-0.22 $[-1.98, 1.54]$
252	-3.94 $[-6.50, -1.39]$	-2.45 $[-3.45, -1.45]$	-1.49 $[-3.69, 0.70]$
503	-5.91 $[-8.91, -2.90]$	-1.42 $[-2.43, -0.41]$	-4.49 $[-7.24, -1.74]$
1,006	-10.44 $[-13.49, -7.39]$	-0.18 $[-1.10, 0.74]$	-10.26 $[-13.20, -7.32]$
2,013	-13.27 $[-16.28, -10.26]$	$+0.41$ $[-0.50, 1.31]$	-13.68 $[-16.68, -10.67]$

At 2,013 km, anomaly energy is lower in 16 of 33 summers, whereas the alignment component is negative in all 33. The corresponding counts at 1,006 km are 21 and 28. The broad-scale contraction is therefore associated with daily anomalies opposing the persistent monthly field, while the transient anomaly energy itself changes little. The decomposition is descriptive and conditional on the chosen climatology. The maximum aggregated identity error over the five graphs is 1.82×10^{-16}.

As a sensitivity analysis, the monthly climatology for each evaluated summer was recomputed from the other 34 summers. Table S5 compares this leave-one-summer-out construction with the inclusive 35-summer climatology. The broad-scale accounting is practically unchanged: at $h = 2{,}013$ km the anomaly-energy component is $+0.40$ rather than $+0.41$ percentage points, and the signed cross component is -13.67 rather than -13.68 points.

Table S5. Sensitivity of the exact decomposition to leave-one-summer-out monthly climatology. Entries are percentage points. "Cross" denotes the signed climatology–anomaly bilinear term.

h (km)	Inclusive anomaly	Inclusive cross	LOSO anomaly	LOSO cross
126	-2.63	-0.22	-2.65	-0.20
252	-2.45	-1.49	-2.47	-1.47
503	-1.42	-4.49	-1.44	-4.47
1,006	-0.18	-10.26	-0.19	-10.25
2,013	$+0.41$	-13.68	$+0.40$	-13.67

S5.2. Spatial projections

For a centred basis matrix $\mathbf{B}$, define the graph-orthogonal projection

$$\widehat{\boldsymbol{\beta}}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^- \mathbf{B}^\top \mathbf{L}_h \mathbf{y}_t, \qquad \widehat{\mathbf{y}}_{th} = \mathbf{B}\widehat{\boldsymbol{\beta}}_{th}.$$

The structured energy of $\widehat{\mathbf{y}}_{th}$ and the residual energy of $\mathbf{y}_t - \widehat{\mathbf{y}}_{th}$ add exactly to $Q_h(\mathbf{y}_t)$. Table S6 compares centred latitude, the latitude–longitude plane and a latitude–longitude–elevation basis. Elevation comes from the invariant ERA5-Land geopotential field: the nearest 0.1-degree grid value at each analysis site was divided by $9.80665\ \mathrm{m\,s^{-2}}$. The 121 elevations range from -0.9 to $1{,}940.3$ m. Each basis column was centred and scaled before the projection. Such scaling leaves its column space unchanged.

Table S6. Nested spatial-basis comparison. Values are mean percentage-point contributions across the evaluation summers. The structured and residual columns for each basis add to the total, apart from rounding.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Geographic–topographic structured	Geographic–topographic residual
126	-2.86	-2.58	-0.28	-2.81	-0.04	-1.16	-1.69
252	-3.94	-4.60	$+0.66$	-5.07	$+1.13$	-2.85	-1.10
503	-5.91	-8.38	$+2.48$	-9.32	$+3.42$	-6.13	$+0.23$
1,006	-10.44	-12.73	$+2.29$	-13.94	$+3.50$	-11.39	$+0.94$
2,013	-13.27	-14.96	$+1.69$	-16.23	$+2.96$	-14.34	$+1.07$

The latitude component is negative in 33 of 33 summers at every scale. The planar component is negative in 32 summers at 126 km and all 33 summers at the other four scales. At 2,013 km, its 95% descriptive (t)-reference interval based on between-summer variation is $[-18.16, -14.30]$ points, and the residual interval is $[1.59, 4.33]$ points. Adding longitude makes the structured component more negative and the compensating residual more positive.

The geographic–topographic component is negative in 18, 21, 28, 33 and 33 of the 33 summers as bandwidth increases. At 126 km its estimate is -1.16 points with a 95% descriptive (t)-reference interval based on between-summer variation of $[-2.66, 0.33]$, which includes zero. At 2,013 km the estimate is -14.34 points $[-16.75, -11.92]$, and the residual is $+1.07$ points. The broad contraction persists after adding elevation to the latitude–longitude basis. The projection yields a combined geographic–topographic contribution; no order-invariant elevation contribution or physical mechanism follows from it. The maximum day-level identity error for the augmented basis is 8.88×10^{-14}, and the maximum aggregated error is 2.78×10^{-16}.

S6. Sensitivity to spatial specification

S6.1. The bandwidth profile

The estimand is the equal-weight mean over five design bandwidths. A denser curve evaluates 31 logarithmically spaced values between 125.800 and 2,012.796 km on the same 121 sites, using the evaluation fields and labels. The five design bandwidths define the estimand; the denser curve describes its scale profile.

All 30 adjacent changes are negative, and all 31 pointwise 95% descriptive (t)-reference intervals based on between-summer variation have upper endpoints below zero. The five design estimates remain $-2.86, -3.94, -5.91, -10.44$ and -13.27%. The curve shows no reversal at the 31 evaluated bandwidths. The intervals are descriptive pointwise summaries; no simultaneous confidence band is estimated.

Table S7. The 31-point bandwidth curve. Values are mean graph-dispersion effects in per cent. The three column pairs continue the same ordered curve.

h (km)	Effect	h (km)	Effect	h (km)	Effect
125.8	-2.856	347.7	-4.596	876.1	-9.538
138.0	-2.958	381.4	-4.835	961.0	-10.151
151.3	-3.076	418.3	-5.125	1,054.0	-10.720
166.0	-3.211	458.8	-5.479	1,156.0	-11.239
182.1	-3.360	503.2	-5.907	1,268.0	-11.703
199.7	-3.521	551.9	-6.407	1,390.8	-12.113
219.0	-3.688	605.4	-6.974	1,525.4	-12.470
240.2	-3.859	664.0	-7.591	1,673.1	-12.778
263.5	-4.031	728.3	-8.239	1,835.1	-13.043
289.0	-4.206	798.8	-8.895	2,012.8	-13.268
317.0	-4.391				

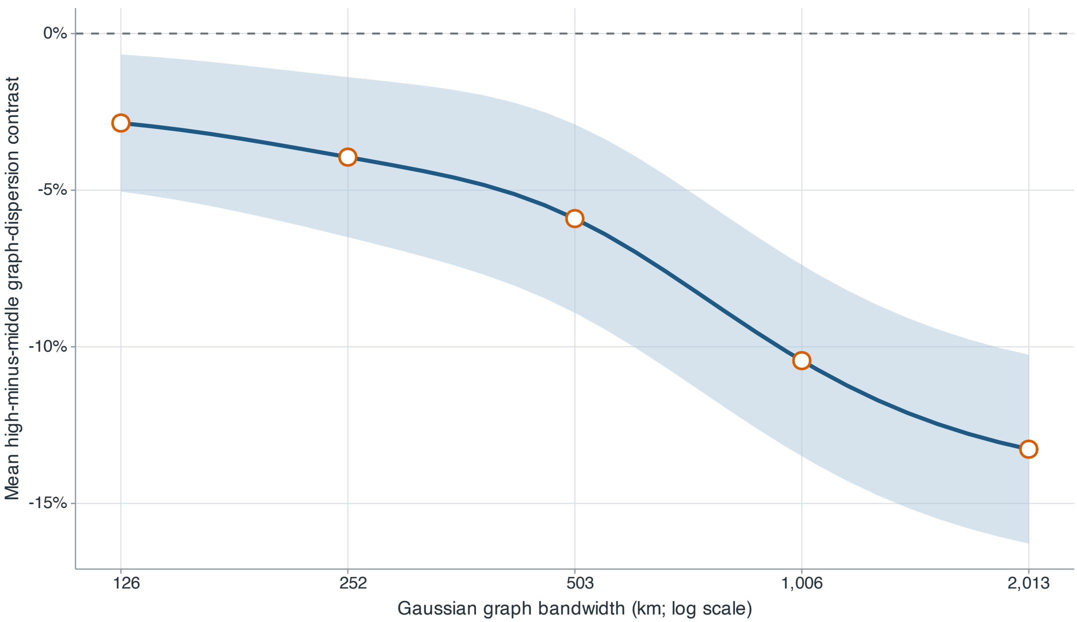

Figure S1 Bandwidth-resolution profile on the 121-site support. The blue line and shaded band show the 31-point profile and its pointwise 95% descriptive (t)-reference intervals based on between-summer variation; orange open circles mark the five design bandwidths.

S6.2. Sampling resolution

The convergence comparison uses the five physical bandwidths, evaluation peak times and 121-site labels. The 239-site configuration refines longitude; the 236-site configuration refines latitude. Each contains all 121 sites and is a subset of the 465-site grid. The two one-direction refinements differ in their added sites.

Table S8. Four-configuration spatial-resolution comparison. Effects and the two error summaries are in percentage points. Differences use the 465-site result at the same physical bandwidth as the reference.

Configuration	Sites	126	252	503	1,006	2,013	Max. $\lvert\Delta\rvert$	RMSE
Analysis support	121	-2.856	-3.945	-5.907	-10.441	-13.268	0.991	0.764
Longitude refined	239	-2.839	-3.846	-5.968	-10.552	-13.351	0.929	0.728
Latitude refined	236	-3.129	-4.308	-6.529	-10.865	-13.458	0.500	0.384
Dense support	465	-3.395	-4.543	-6.898	-11.340	-13.959	0.000	0.000

Every support retains the broad-to-local ordering and a negative effect at all five bandwidths. Relative to 465 sites, the 121-site profile has a maximum absolute discrepancy of 0.99 points and an across-scale RMSE of 0.76 points. These finite-grid comparisons show how the estimates change with spatial support. Agreement is weakest for the narrowest graph, consistent with its shorter length scale relative to intersite spacing.

These convergence calculations use the primary equirectangular distance matrix. On the 465-site grid, the original evaluation events give -8.03%; reselecting events with the same distance matrix gives -7.25%. The WGS84 relabelled calculation for the full 465-site domain, reported below, instead gives -6.93%.

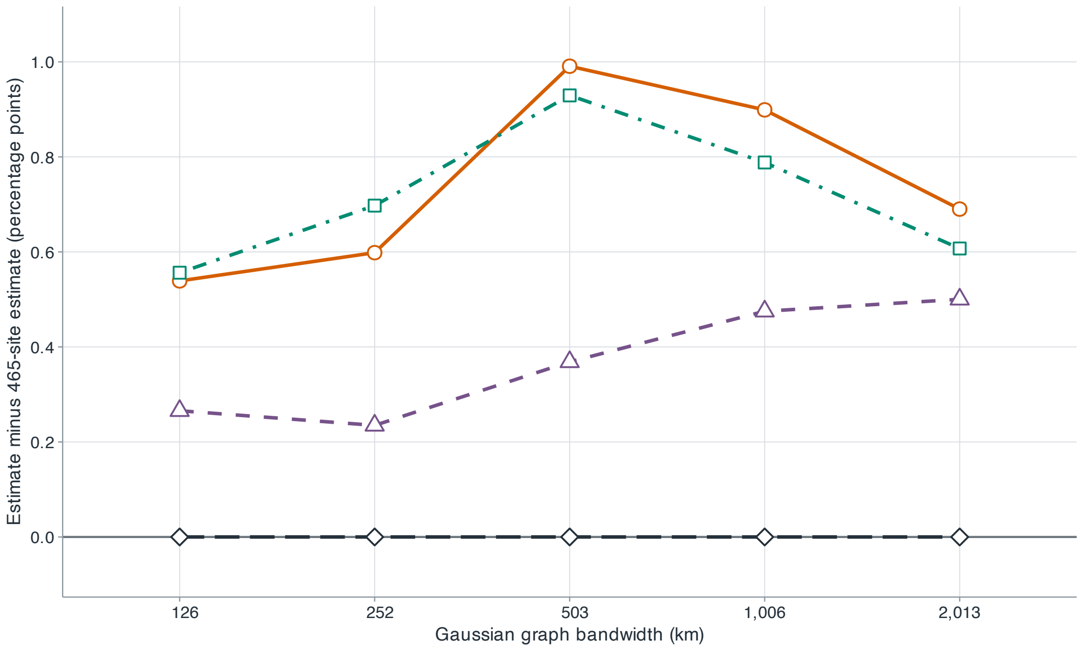

Figure S2 Differences from the 465-site estimate at five bandwidths. Orange circles with a solid line denote the 121-site support; purple triangles with a dashed line, the latitude-refined 236-site support; teal squares with a dot-dash line, the longitude-refined 239-site support; and dark diamonds with a long-dash line, the dense 465-site reference.

S6.3. Geographic specification

Table S9 reports the complete five-bandwidth profiles from the additional support calculations. The 121-site WGS84 rows change the distance calculation and, where stated, the target weights and event labels. The remaining rows use the 465-site downloaded lattice, WGS84 distances, equal site weights and labels recomputed from each retained domain. For compactness, the table reports the larger of the two inward movements at each boundary; the retained results also contain both movements, fixed-label variants, cosine-latitude variants, weighted pair distances, effective edge counts and between-summer summaries.

For the 121-site cosine-latitude target, the primary equirectangular matrix gives -8.11% with original labels and -7.67% after relabelling. The corresponding WGS84 values in Table S9 are -7.80% and -7.37%.

All displayed profiles are negative and strengthen with bandwidth. Relabelled boundary means range from -6.32% to -7.97%; the north boundary has the largest influence on magnitude. The Natural Earth row operationalises a non-rectangular target as the low-resolution feature named China, excluding the separately stored Taiwan feature, intersected with the downloaded lattice and its ERA5-Land-valid mask. These checks support the direction and scale ordering, but they also show that the magnitude belongs to the specified domain.

Table S9. Graph-dispersion contrasts under distance and spatial-support changes. Entries are percentages at the five Gaussian bandwidths; Mean is their equal-weight average. Boundary distances are inward movements because no fields were downloaded outside the original rectangle.

Analysis	Sites	126	252	503	1,006	2,013	Mean
Primary equirectangular, equal-site, fixed labels	121	-2.86	-3.94	-5.91	-10.44	-13.27	-7.28
WGS84, equal-site, fixed labels	121	-2.42	-3.47	-5.50	-10.26	-13.22	-6.97
WGS84, cosine-latitude, fixed labels	121	-3.11	-4.33	-6.45	-11.14	-14.00	-7.80
WGS84, cosine-latitude, relabelled	121	-2.95	-4.04	-6.00	-10.54	-13.31	-7.37
Full downloaded domain, relabelled	465	-2.44	-3.34	-5.61	-10.26	-13.02	-6.93
South edge inward $1.8°$, relabelled	457	-2.28	-2.99	-5.11	-9.62	-12.22	-6.44
North edge inward $1.8°$, relabelled	413	-2.57	-3.91	-7.00	-11.94	-14.41	-7.97
West edge inward $1.6°$, relabelled	415	-1.97	-2.31	-4.55	-9.85	-12.90	-6.32
East edge inward $1.6°$, relabelled	458	-2.44	-3.22	-5.42	-10.07	-12.80	-6.79
Natural Earth China-mainland intersection, relabelled	445	-2.09	-2.80	-5.06	-9.64	-12.18	-6.35

S7. Historical consistency

The ERA5-Land archive distributed through the Copernicus Climate Data Store begins in January 1950.[1] A plan dated 9 August 2026 defined the 41 summers, 121 sites, field construction, five scales, aggregation and product-shift calculation before the historical values were retrieved. The analysis weights the three months, five scales and summers equally. The two periods are separated in calendar time but share the ERA5-Land model and assimilation system. The product-shift calculation uses 99,999 draws, seed 20260810 and joint rotation of all five dispersion columns within each month-year record.

The five-scale contrast over 1950–1990 is -11.04% (descriptive t-reference interval $[-12.84, -9.24]\%$ based on between-summer variation), with a negative estimate in 40 of 41 summers. Deleting one summer at a time gives estimates from -11.33% to -10.62%. Table S10 gives the period split and the five-scale profile.

Table S10. The 1950–1990 historical extension. Estimates and 95% descriptive (t)-reference intervals based on between-summer variation are percentages; the final column counts summers with a negative estimate. Period rows average the five bandwidths. Scale rows use all 41 summers.

Period or bandwidth	Summers	Estimate	95% descriptive (t)-reference interval	Negative summers
1950–1990 overall	41	-11.04	$[-12.84, -9.24]$	40
1950–1978	29	-10.50	$[-12.50, -8.51]$	28
1979–1990	12	-12.34	$[-16.65, -8.04]$	12
126 km	41	-5.96	$[-7.37, -4.54]$	38
252 km	41	-7.51	$[-9.15, -5.87]$	39
503 km	41	-10.08	$[-12.11, -8.04]$	40
1,006 km	41	-14.50	$[-16.72, -12.27]$	41
2,013 km	41	-17.17	$[-19.44, -14.91]$	41

Provider documentation identifies the forcing change used for the period split: ERA5-Land for 1950–1978 used the preliminary ERA5 back extension, whereas the later period used the consolidated ERA5 release.[2] Both period estimates are negative, and their intervals overlap. The 1950–1978 estimate inherits uncertainty from the preliminary forcing data.

We applied the product cyclic-shift calculation to 123 month-year records. None of the sampled statistics was at or below the observed -11.043%, so the plus-one lower-tail value is 0.00001. The sampled null mean and standard deviation are 0.853% and 1.436 percentage points. Every scale has an unadjusted value of 0.00001; the complete-null single-step values range from 0.00001 to 0.00006.

[1] Dataset overview: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview.

[2] ERA5-Land production note: https://confluence.ecmwf.int/pages/viewpage.action?pageId=402639006.

For the historical climatology–anomaly decomposition, the monthly reference field was estimated within 1950–1990:

$$\boldsymbol{\mu}_m^H = \frac{1}{41}\sum_{y=1950}^{1990}\left(\frac{1}{n_m}\sum_{t\in m}\mathbf{y}_{yt}\right), \qquad \mathbf{a}_{yt}^H = \mathbf{y}_{yt} - \boldsymbol{\mu}_m^H.$$

Thus $\boldsymbol{\mu}_m^H$ is the site-specific mean field for month m across the 41 historical summers. The anomaly and alignment terms therefore describe departures from the historical period's own persistent monthly spatial pattern.

Table S11 reports the latitude, planar and climatology–anomaly accounting for the historical extension. The latitude and planar structured components are negative in all 41 summers at every bandwidth. At 2,013 km, anomaly energy is -0.28 points with interval $[-1.01, 0.45]$, while the climatology–anomaly alignment component is -16.89 points $[-19.31, -14.47]$ and is negative in all 41 summers. The broad-scale historical result is therefore also alignment-dominated, and its latitude and planar components have the same negative direction as in the 1991–2025 period.

Table S11. Structural accounting for the 1950–1990 extension. Entries are percentage points. Within each geographic basis, the structured and residual columns add to the total. The anomaly and alignment columns also add to the total because the climatology-energy difference is zero.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Anomaly energy	Climatology–anomaly alignment
126	-5.96	-2.83	-3.13	-3.15	-2.81	-3.22	-2.74
252	-7.51	-4.92	-2.59	-5.50	-2.01	-3.30	-4.21
503	-10.08	-8.86	-1.21	-9.92	-0.15	-2.41	-7.67
1,006	-14.50	-13.60	-0.90	-14.99	$+0.49$	-1.02	-13.48
2,013	-17.17	-16.13	-1.05	-17.62	$+0.44$	-0.28	-16.89

S8. NOAA station comparison

The station comparison uses the ten summers enumerated before station retrieval: 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025. The set is $\{1992 + 4k : k = 0, \ldots, 7\} \cup \{2023, 2025\}$. The official Integrated Surface Database (ISD) station-history snapshot used for selection was archived.[3] The planned eligibility rule required a station-history end date on or after 31 August 2025, but that snapshot contained no such candidate: its global maximum end date was 28 August and its maximum inside the study rectangle was 24 August. The operational administrative cutoff was therefore 24 August. This administrative change preceded retrieval of station WBT observations and matched ERA5-Land values.

For a candidate station s, the proximity criterion was

$$d(s, \mathcal{S}_{121}) = \min_{i\in\mathcal{S}_{121}} d(s,i) \leq 150 \text{ km},$$

where $\mathcal{S}_{121}$ is the 121-site analysis support. For coordinate differences in degrees, the pairwise distance was

$$d(s,i) = \left[\{111.32\cos(\bar{\phi}_{si})(\lambda_s - \lambda_i)\}^2 + \{110.57(\phi_s - \phi_i)\}^2\right]^{1/2},$$

with $\bar{\phi}_{si}$, the pair's mean latitude, converted to radians in the cosine. This filter was applied before maximin selection and restricted station locations to within 150 km of the analysis support. Of the 228 stations satisfying the rectangle and date conditions, 20 exceeded 150 km, leaving 208 candidates; the largest retained nearest-site distance was 147.73 km and the smallest excluded distance was 155.72 km. A deterministic maximin rule based on station coordinates selected 30 stations. The study rectangle defined the candidate domain. A station-year qualified with

[3] NOAA Integrated Surface Database: https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database.

at least 400 exact UTC hours having temperature, dew point and usable station pressure; 26–30 of the 30 selected stations qualified in each year. Station wet-bulb temperature was calculated with the same Bolton implementation used for ERA5-Land. ISD sea-level pressure p_{SL} was converted to station pressure using the standard-atmosphere relation

$$p_s = 100 p_{\mathrm{SL}} \max\{1 - 2.25577 \times 10^{-5} z, 0.1\}^{5.2559},$$

where p_{SL} is in hPa and elevation z is in metres. Selected station elevations ranged from 3 to 2,063 m. Treating sea-level pressure as surface pressure instead changed WBT by 0.072 °C on average in the 175,172 matched hours; the maximum absolute difference was 1.346 °C.

The nearest ERA5-Land cells for the Shengsi and Taidong island stations had no finite land values at any requested time, so both point series were recorded as unavailable. The other 28 station locations had finite ERA5-Land series. Combining this land-mask restriction with the hourly qualification rule left 24–28 matched stations per year and 175,172 paired hours. Table S12 reports the measurement comparison. Bias is ERA5-Land minus station wet-bulb temperature; the centred correlation subtracts each station mean before pooling. The all-year row pools all matched hours across years.

Table S12. ERA5-Land and NOAA station wet-bulb agreement in ten evaluation years. Bias, MAE and RMSE are in °C. Stations are the qualified locations with a finite nearest-cell ERA5-Land series in that year.

Year	Stations	Matched hours	Bias	MAE	RMSE	Centred correlation
1992	25	17,887	-0.25	0.88	1.16	0.919
1996	25	19,147	-0.33	0.88	1.16	0.908
2000	25	16,765	-0.40	0.94	1.21	0.908
2004	25	16,829	-0.20	0.91	1.18	0.923
2008	25	17,310	-0.20	0.88	1.14	0.909
2012	25	17,349	-0.27	0.93	1.22	0.918
2016	24	16,589	-0.34	0.88	1.13	0.937
2020	28	19,323	-0.26	1.04	1.48	0.897
2023	27	16,554	-0.19	1.02	1.49	0.925
2025	28	17,419	-0.17	1.01	1.45	0.933
All years	28	175,172	-0.26	0.94	1.27	0.916

For the effect comparison, ERA5-Land peak times and high/middle labels defined the event days, and we aligned station observations to those events. Each graph field used the same available station subset for NOAA and ERA5-Land and required at least ten stations. All 30 month-year records retained at least one high and one middle field, giving 345 usable event fields. We formed high-to-middle ratios within records, then averaged them equally over months within years and over the ten years. Table S13 reports 95% descriptive (t)-reference intervals based on between-summer variation. Inference focuses on the 503-, 1,006- and 2,013-km graphs supported by the station spacing; the two narrower graphs remain descriptive.

Availability was nearly the same by ERA5-Land-defined state: 122 of 240 scheduled high fields (50.8%) and 223 of 440 middle fields (50.7%) had at least ten stations. Their mean available-station counts over all scheduled fields were 13.28 and 13.12. Nevertheless, station support varied over time, and a retained record could have only one field in either state.

Table S13. High–middle graph-dispersion contrasts at ERA5-Land-defined peak times and labels in ten evaluation years. Contrasts and intervals are percentages. "Negative" counts years with a negative scale-specific estimate. Each scale contains all 30 month-year records and an average of 25.3 stations per event field.

Bandwidth (km)	NOAA contrast [95% descriptive (t)-reference interval]	NOAA negative years	ERA5-Land contrast [95% descriptive (t)-reference interval]	ERA negative years
126	$-6.23\ [-25.87, 13.41]$	6/10	$-11.60\ [-27.49, 4.29]$	7/10
252	$-7.23\ [-21.01, 6.55]$	7/10	$-12.74\ [-22.68, -2.81]$	8/10
503	$-12.50\ [-24.05, -0.96]$	8/10	$-17.13\ [-25.15, -9.11]$	10/10
1,006	$-18.03\ [-29.17, -6.90]$	9/10	$-21.60\ [-29.07, -14.14]$	10/10
2,013	$-21.47\ [-32.51, -10.43]$	9/10	$-24.23\ [-31.72, -16.74]$	10/10

The station and paired ERA5-Land profiles are negative at all five scales and become more negative with bandwidth. Averaging the three station-supported scales within each year gives -17.34% (95% descriptive (t)-reference interval $[-28.49, -6.18]\%$), with a negative average in nine of ten years. The same broad-profile calculation for the matched ERA5-Land fields gives -20.99% ($[-28.57, -13.40]\%$), negative in all ten years. The two fields are compared on identical, time-varying station supports. Because ERA5-Land defines the event times and labels, the comparison measures agreement for common events on the station network. Coverage is weakest at 126 and 252 km. Requiring two, three or five retained days per state gave broad-scale station summaries of -13.42%, -15.24% and -14.56%; the last used only six years. A year-specific fixed common-station support gave -18.08%. Defining daily peaks and high/middle labels from station regional means, with at least five days per state, gave -15.99%. The broad-scale contrast remains negative under these event and station-support definitions.

S9. Contrasts within monthly records

Two alternative event contrasts use the same sites, graphs and aggregation hierarchy. The continuous calculation regresses $\log Q_h$ on centred regional wet-bulb temperature separately in each month-year record, then averages the slopes equally over months, scales and the 33 evaluation summers. The five-scale mean slope is $-0.0650\ ^\circ\mathrm{C}^{-1}$, with a 95% descriptive (t)-reference interval based on between-summer variation of $[-0.0822, -0.0479]$ and negative summer slopes in 31 of 33 years.

The calendar-matched calculation compares each high day with the mean of middle days no more than three or five calendar days away in the same month-year record. Eligible high-day contrasts are averaged within records before application of the original equal-month, equal-scale and equal-summer hierarchy. The ± 3-day window matches 670 of 792 high days (84.6%); the ± 5-day window matches 750 (94.7%). Each of the 99 evaluation month-year records contributes at least one match under both windows.

Table S14. Continuous-intensity and calendar-matched results for the 33 evaluation summers. Continuous entries are slopes of $\log Q_h$ per $^\circ$C. Matching entries are percentage effects. Brackets give 95% descriptive (t)-reference intervals based on between-summer variation.

Bandwidth	Continuous slope [interval]	± 3-day match [interval]	± 5-day match [interval]
126	-0.0318 $[-0.0435, -0.0201]$	-2.32 $[-4.73, 0.09]$	-2.54 $[-4.90, -0.18]$
252	-0.0408 $[-0.0553, -0.0263]$	-3.29 $[-6.03, -0.56]$	-3.54 $[-6.18, -0.90]$
503	-0.0546 $[-0.0735, -0.0356]$	-4.41 $[-7.44, -1.38]$	-4.74 $[-7.66, -1.83]$
1,006	-0.0877 $[-0.1092, -0.0661]$	-7.17 $[-10.32, -4.02]$	-7.90 $[-10.94, -4.86]$
2,013	-0.1104 $[-0.1325, -0.0882]$	-8.87 $[-12.06, -5.68]$	-9.89 $[-12.98, -6.80]$
Five-scale mean	-0.0650 $[-0.0822, -0.0479]$	-5.21 $[-8.01, -2.41]$	-5.72 $[-8.43, -3.02]$

The matched effects are weaker than the unrestricted high-to-middle contrast, but their broad-to-local ordering remains. Calendar matching also changes the target population of day pairs: only high days with eligible middle days enter the high-day average.

S10. Ratio estimation with heavy tails

We compare the raw effect $r-1$, the log effect $\log r$ and the bounded symmetric effect $2(r-1)/(r+1)$, where r is the high-to-middle dispersion ratio. The primary summary averages the raw ratios across records. Small middle-day denominators amplify the upper tail of this asymmetric measure. Its randomisation distribution need not be centred at zero. The 495 observed middle-day denominators range from 2.25 to $25.99\ ^\circ\mathrm{C}^2$; the log ratio is the principal robustness summary. In the observed 33-summer record, the five-scale raw-ratio summary is -7.28%, the back-transformed log-ratio summary is -8.30%, and the bounded symmetric summary is -8.59%.

The stress design uses the 121 observed site locations, exponential spatial covariance with 450-km range and 5% nugget, circular field correlation 0.45 and regional-mean autoregression 0.65. A common daily scale $\sqrt{3/\chi^2_3}$ produces heavy-tailed graph energy. Each scenario has 2,000 paired replications of six 30–31-day records. All three measures use the same 999 product shifts within a replication; the seed is 20260811.

Table S15. Raw, log and bounded effect measures under the heavy-tailed stress design. Bias and RMSE are on each measure's scale. Size and power are lower-tail rejection proportions at 0.05. Alternative targets correspond to a high-day energy multiplier of 0.93.

Measure	Null bias	Null RMSE	Size	Alternative target	Alt. bias	Alt. RMSE	Power
Raw ratio	0.3716	1.4725	0.0470	-0.0700	0.3456	1.3694	0.0795
Log ratio	-0.0698	0.3283	0.0490	-0.0726	-0.0698	0.3283	0.0850
Bounded symmetric	-0.0657	0.2771	0.0475	-0.0725	-0.0574	0.2748	0.0790

The three product-shift tests retain near-nominal size in this design. The raw ratio nevertheless has an RMSE more than five times that of the bounded measure under the null. Product-shift calibration concerns the randomisation distribution of a fixed statistic; ratio-estimation stability also depends on the right tail of daily energy. The log and bounded summaries reduce that instability, with the bounded measure giving the smallest RMSE in these cells. The smaller RMSE of the log and bounded contrasts supports their use when daily energies have heavy right tails.

S11. Simulation specifications

S11.1. Models for daily spatial profiles

The baseline experiment uses the 121 observed coordinates and the five design graphs. Six records contain 30 or 31 days unless a null cell changes the record count or length. Regional means follow an AR(1) process with coefficient 0.65; type-7 within-record quartiles define high and middle days. Centred spatial anomalies have exponential covariance with 450-km range and 5% nugget, and temporal coefficient 0.45 unless changed below. Projection through $\mathbf{I} - \mathbf{1}\mathbf{1}^\top/121$ makes the anomaly mean exactly zero, so the field mean equals the series used for labels. The circular Gaussian nulls use wrap-around temporal covariance and satisfy record-wise cyclic invariance. The two finite-AR cells deliberately omit wrap-around and assess a boundary departure.

The alternatives condition their parameter changes on the high-day indicator. Amplitude multipliers are 0.97, 0.955, 0.94, 0.92 and 0.90. The archived range grid is labelled 525, 587, 650, 725 and 800 km, but the simulator uses three covariance matrices: both settings below 600 km use a 525-km range, both settings from 600 to below 750 km use a 650-km range, and the last uses 800 km. Table S17 reports all 25 simulation cells; the five range labels correspond to three distinct covariance designs. The range change enters the innovation covariance and can persist through the field AR(1) state. Gradient alternatives add the standardised pattern proportional to $0.8x + y$ with middle-day amplitude 1.8 and high-day amplitudes 1.6, 1.5, 1.4, 1.3 and 1.2. Conditional targets are computed from the known quadratic forms rather than from Monte Carlo averages.

Seven tests use the same simulated fields, labels and sampled shifts. The graph test averages the five Gaussian-graph ratios; the variogram test averages five pair-distance-quintile ratios. The remaining tests use spatial variance, four-nearest-neighbour semivariance, 400–600-km binned semivariance, Moran's I and Geary's C. The first five are lower-tail tests; Moran and Geary are two-sided. Each of the 25 archived cells, including the two duplicated range implementations, contains 1,000 replications and 499 shifts. With 1,000 replications, the Monte Carlo standard error is 0.69 percentage points at 5% rejection and 1.58 points at 50% power.

The graph profile has the largest minimum power across the 15 alternatives, but nearest-neighbour semivariance is stronger for range expansion and variance or the variogram profile is stronger for weak gradient suppression. Standardisation leaves Moran's I and Geary's C with near-null power for pure amplitude changes.

Table S16. Complete null calibration for the baseline day-level experiment. Entries are rejection percentages at 0.05 over 1,000 replications.

Null cell	Graph	Variogram	Variance	Nearest	400–600 km	Moran	Geary
Circular baseline	4.6	4.9	5.4	5.5	4.3	4.7	5.2
Circular, 3 records	6.8	6.1	6.0	5.8	5.2	5.7	5.0
Circular, 12 records	5.3	5.7	5.9	4.4	4.8	4.9	4.2
Circular, 20 days	5.8	5.3	5.0	5.2	6.4	4.8	4.8
Circular, 60 days	6.2	5.7	5.9	5.9	6.0	6.1	6.1
Circular, $\rho = 0$	6.2	5.8	5.6	4.0	5.7	5.6	5.2
Circular, $\rho = 0.75$	4.5	4.2	4.1	5.4	4.1	4.9	4.1
Circular, t_3 scale	6.1	5.6	5.6	6.1	5.9	5.0	6.1
Finite AR baseline	4.6	4.2	4.4	4.6	4.6	5.5	4.9
Finite AR, $\rho = 0.75$	5.1	5.1	5.0	4.6	5.1	4.7	5.8

Table S17. Complete power results for the baseline day-level alternatives. Target is the true five-graph contraction in per cent; other entries are rejection percentages at 0.05 over 1,000 replications.

Mechanism	Setting	Target	Graph	Variogram	Variance	Nearest	400–600 km	Moran	Geary
Amplitude	0.970	-5.91	50.0	33.6	31.1	76.4	45.0	5.2	5.9
Amplitude	0.955	-8.80	80.5	57.3	54.4	96.9	75.1	5.0	5.0
Amplitude	0.940	-11.64	95.3	79.8	77.1	99.8	91.9	4.6	4.5
Amplitude	0.920	-15.36	99.7	93.3	91.1	100.0	99.0	4.4	3.8
Amplitude	0.900	-19.00	99.9	99.3	99.1	100.0	99.9	5.4	4.7
Range	525 km	-6.07	48.9	19.0	16.4	95.4	51.5	15.5	15.9
Range	587-km archived label (implemented as 525 km)	-6.07	50.8	20.3	17.8	95.2	52.4	13.2	15.1
Range	650 km	-14.38	98.6	62.4	53.3	100.0	99.1	45.3	53.7
Range	725-km archived label (implemented as 650 km)	-14.39	98.8	60.8	51.0	100.0	99.6	49.0	56.2
Range	800 km	-22.20	100.0	91.4	85.6	100.0	100.0	74.0	85.0
Gradient	1.6	-10.34	90.9	93.9	96.5	40.8	78.9	32.4	81.6
Gradient	1.5	-15.06	99.6	99.9	99.9	65.2	95.0	64.8	98.7
Gradient	1.4	-19.47	100.0	100.0	100.0	82.5	99.8	87.6	100.0
Gradient	1.3	-23.57	100.0	100.0	100.0	94.5	100.0	98.0	100.0
Gradient	1.2	-27.37	100.0	100.0	100.0	97.6	100.0	99.8	100.0

S11.2. Joint generation of fields and event labels

The coupled simulation uses the 121 observed site coordinates and six records of lengths 30, 31, 30, 30, 31 and 30 days. Isotropic anomaly fields have exponential covariance with 450-km range and 5% nugget. The field state has autoregressive coefficient 0.45, and the regional weather series has coefficient 0.65. High and middle labels use type-7 quartiles within each record. A standardised spatial gradient is proportional to $0.8x + y$, where x and y are projected site coordinates. Its sign is driven by $0.75\widetilde{\mu}_t + (1 - 0.75^2)^{1/2} Z_t$, so labels and spatial structure share a latent weather component even under the null.

Table S18. Exact changes defining the eight coupled label–field designs. Every cell retains the common six-record construction and within-record labels.

Design	Null	Contraction alternative
Shared latent gradient	Gradient amplitude is 1.8 on every day; its sign shares the latent driver with regional weather.	Gradient amplitude on high days is reduced from 1.8 to 1.2.
Within-month progression	Regional mean is a linear trend from -1.4 to 1.4 plus 0.60 times the AR(1) weather series; gradient amplitude remains 1.8.	Gradient amplitude declines linearly from 1.95 to 1.30 across the record.
Anisotropic covariance	Anomaly e-folding ranges are 800 km in projected longitude and 300 km in projected latitude.	High-day anomaly amplitude is multiplied by 0.94.
Peak-hour selection	Each daily regional mean is the maximum of 24 hourly values with sinusoidal amplitude 1.8 and independent noise SD 0.35; anomaly covariance is hour-invariant.	High-day anomaly amplitude is multiplied by 0.94.

Each field is evaluated with the five Gaussian graphs and a five-bin variogram profile whose bins are the quintiles of the 7,260 observed pair distances. For each replication, the target is the high-to-middle contrast in the conditional expected quadratic forms. The graph and variogram statistics use the same simulated fields, labels and product shifts. Each of the eight cells contains 1,000 replications with 499 shifts per test.

S11.3. Repeated sampling of summers

The annual experiment crosses $R \in \{20, 33, 60, 120\}$, $\rho \in \{0, 0.3, 0.6\}$, Gaussian, centred log-normal and standardised t_3 innovations, and effects $\delta \in \{0, -0.035, -0.07\}$. The resulting 108 cells each contain 10,000 replications after a 200-step burn-in. For retained summer r,

$$U_r = \rho U_{r-1} + (1-\rho^2)^{1/2}\varepsilon_r, \qquad Y_r = \delta + 0.12U_r.$$

The Gaussian innovation is standard normal. For $Z \sim N(0,1)$, the centred unit-variance log-normal innovation is

$$\frac{\exp(0.65Z) - \exp(0.65^2/2)}{\{[\exp(0.65^2) - 1]\exp(0.65^2)\}^{1/2}},$$

and the heavy-tailed innovation is $T_3/\sqrt{3}$. The point estimate is $\bar{Y}$. The ordinary interval uses the sample variance and a t_{R-1} critical value. The lag-2 calculation uses a Bartlett long-run variance with the same critical value. The growing-lag interval uses $L = \max\{1, \lfloor R^{1/3} \rfloor\}$ and a standard-normal critical value. The sign test uses the exact binomial upper tail of the number of negative summers. The sign calculation targets $\Pr(Y_r < 0) = 1/2$. The mean-null target is $\mathrm{E}(Y_r) = 0$. Under the centred skewed innovations used below, the two targets can differ; sign-column rates in skewed mean-null cells therefore quantify rejection of the sign null. The three-test decision requires the Student, lag-2 and sign p-values all to be at most 0.025. Standardised diagnostics use the known long-run variance $0.12^2(1+\rho)/(1-\rho)$.

The following tables report all eight coupled cells and all 108 annual cells. Machine-readable versions of all 108 annual cells are provided in `results/year_inference_simulation_summary.csv` in the accompanying repository.

Table S19. Complete results for the eight coupled label–field stress designs. Targets, bias and RMSE are percentage points; rejection is per cent over 1,000 replications. Graph and five-bin variogram profiles use identical fields, labels and product shifts.

Scenario	Family	Graph profile				Five-bin variogram			
		Target	Bias	RMSE	Reject	Target	Bias	RMSE	Reject
Shared-latent null	Null	0.00	0.38	3.63	5.5	-0.00	0.53	4.58	5.3
Shared-latent contraction	Alternative	-27.37	0.19	2.70	100.0	-36.66	0.25	3.26	100.0
Seasonal-progression null	Null	0.00	0.64	3.76	4.0	-0.00	0.85	4.77	4.3
Seasonal-gradient contraction	Alternative	-10.74	0.27	3.65	78.6	-14.72	0.33	4.60	83.3
Anisotropic null	Null	0.00	0.31	3.86	7.0	0.00	0.42	4.81	7.3
Anisotropic amplitude contraction	Alternative	-5.92	0.21	3.63	51.4	-3.98	0.32	4.58	25.5
Peak-selection null	Null	0.00	0.21	3.69	6.6	-0.00	0.34	4.63	6.6
Peak-selection contraction	Alternative	-5.90	0.36	3.50	54.2	-3.96	0.52	4.46	25.1

Table S20. Simulation results for $R = 20$ summers: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
				percentage points		coverage (%)		
20	0.0	Gaussian	Null	0.015	2.655	95.4	91.7	89.9
20	0.0	Gaussian	-3.5%	-0.015	2.680	94.8	91.1	89.2
20	0.0	Gaussian	-7%	-0.015	2.665	95.1	91.3	89.4
20	0.0	Skewed	Null	-0.046	2.691	91.2	87.6	85.9
20	0.0	Skewed	-3.5%	0.022	2.676	92.1	88.7	86.9
20	0.0	Skewed	-7%	-0.024	2.662	91.8	88.3	86.5
20	0.0	t_3	Null	-0.028	2.672	95.3	91.7	89.7
20	0.0	t_3	-3.5%	0.028	2.599	95.9	92.0	90.4
20	0.0	t_3	-7%	0.028	2.696	95.9	92.0	90.1
20	0.3	Gaussian	Null	0.081	3.584	85.5	86.8	84.8
20	0.3	Gaussian	-3.5%	-0.096	3.597	85.0	86.6	84.2
20	0.3	Gaussian	-7%	0.036	3.620	85.0	86.5	84.0
20	0.3	Skewed	Null	0.053	3.612	82.7	84.4	82.3
20	0.3	Skewed	-3.5%	0.019	3.615	82.6	84.2	82.2
20	0.3	Skewed	-7%	-0.038	3.545	83.0	84.6	82.3
20	0.3	t_3	Null	0.056	3.486	85.9	87.5	85.1
20	0.3	t_3	-3.5%	-0.126	3.533	85.7	87.3	84.8
20	0.3	t_3	-7%	-0.002	3.620	85.8	87.8	85.3
20	0.6	Gaussian	Null	-0.042	5.096	66.5	76.2	73.4
20	0.6	Gaussian	-3.5%	0.002	5.128	67.0	76.3	73.8
20	0.6	Gaussian	-7%	0.000	5.175	66.6	75.9	73.2
20	0.6	Skewed	Null	0.038	5.187	64.4	73.9	71.2
20	0.6	Skewed	-3.5%	-0.043	5.140	64.5	74.3	71.5
20	0.6	Skewed	-7%	-0.039	5.115	65.1	74.6	71.7
20	0.6	t_3	Null	0.047	4.987	65.9	76.3	73.5
20	0.6	t_3	-3.5%	0.075	5.101	65.1	75.8	72.7
20	0.6	t_3	-7%	-0.026	5.254	65.5	76.7	73.7

Table S21. Simulation results for $R = 33$ summers: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
				percentage points			coverage (%)	
33	0.0	Gaussian	Null	0.057	2.065	95.2	93.3	91.2
33	0.0	Gaussian	-3.5%	-0.004	2.078	95.2	93.1	91.2
33	0.0	Gaussian	-7%	-0.021	2.062	95.2	93.5	91.4
33	0.0	Skewed	Null	-0.029	2.071	92.4	90.6	88.5
33	0.0	Skewed	-3.5%	-0.002	2.061	92.9	91.0	89.0
33	0.0	Skewed	-7%	0.005	2.080	92.6	90.7	88.8
33	0.0	t_3	Null	-0.019	2.089	95.8	93.7	91.5
33	0.0	t_3	-3.5%	0.007	2.062	95.6	93.2	91.0
33	0.0	t_3	-7%	-0.004	2.127	95.7	93.6	91.5
33	0.3	Gaussian	Null	-0.046	2.837	85.4	89.0	87.3
33	0.3	Gaussian	-3.5%	-0.002	2.819	85.4	88.6	87.2
33	0.3	Gaussian	-7%	0.018	2.808	85.7	88.8	87.3
33	0.3	Skewed	Null	0.000	2.795	83.3	87.2	85.9
33	0.3	Skewed	-3.5%	-0.004	2.834	82.9	86.4	85.1
33	0.3	Skewed	-7%	-0.008	2.792	83.4	86.8	85.4
33	0.3	t_3	Null	-0.011	2.773	85.0	89.2	87.5
33	0.3	t_3	-3.5%	0.012	2.804	85.5	89.4	88.2
33	0.3	t_3	-7%	0.033	2.760	85.4	89.1	87.8
33	0.6	Gaussian	Null	0.004	4.112	66.1	78.7	79.0
33	0.6	Gaussian	-3.5%	0.031	4.059	66.8	79.1	79.2
33	0.6	Gaussian	-7%	0.027	4.049	67.3	79.5	79.6
33	0.6	Skewed	Null	0.020	4.090	64.9	77.6	77.9
33	0.6	Skewed	-3.5%	0.043	4.053	65.6	77.8	78.0
33	0.6	Skewed	-7%	0.037	4.070	65.0	77.7	77.7
33	0.6	t_3	Null	-0.014	3.994	66.5	79.8	80.1
33	0.6	t_3	-3.5%	-0.054	4.004	66.0	79.9	80.3
33	0.6	t_3	-7%	0.006	4.158	65.4	79.6	79.8

Table S22. Simulation results for $R = 60$ summers: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
				percentage points			coverage (%)	
60	0.0	Gaussian	Null	0.002	1.554	95.1	94.0	92.9
60	0.0	Gaussian	-3.5%	-0.018	1.562	94.8	93.6	92.4
60	0.0	Gaussian	-7%	-0.003	1.551	95.0	93.8	92.8
60	0.0	Skewed	Null	-0.002	1.553	93.4	92.2	91.1
60	0.0	Skewed	-3.5%	0.020	1.541	93.5	92.5	91.4
60	0.0	Skewed	-7%	0.039	1.545	94.1	93.2	92.0
60	0.0	t_3	Null	0.019	1.583	95.4	94.3	93.2
60	0.0	t_3	-3.5%	0.008	1.523	95.5	94.3	93.3
60	0.0	t_3	-7%	-0.008	1.536	95.7	94.8	93.6
60	0.3	Gaussian	Null	0.016	2.097	85.0	90.0	89.7
60	0.3	Gaussian	-3.5%	-0.016	2.068	85.9	90.9	90.4
60	0.3	Gaussian	-7%	0.033	2.083	85.6	90.6	90.3
60	0.3	Skewed	Null	-0.019	2.085	84.2	89.0	88.8
60	0.3	Skewed	-3.5%	0.018	2.096	84.1	89.1	88.8
60	0.3	Skewed	-7%	-0.001	2.097	83.7	88.6	88.4
60	0.3	t_3	Null	-0.014	2.082	85.2	90.5	90.0
60	0.3	t_3	-3.5%	0.035	2.088	84.7	90.7	90.3
60	0.3	t_3	-7%	-0.010	2.087	85.1	90.5	90.1
60	0.6	Gaussian	Null	-0.011	3.056	67.0	81.0	82.5
60	0.6	Gaussian	-3.5%	-0.029	3.066	66.9	81.5	83.0
60	0.6	Gaussian	-7%	0.030	3.036	67.3	81.3	82.8
60	0.6	Skewed	Null	0.009	3.026	65.9	80.1	81.5
60	0.6	Skewed	-3.5%	-0.016	3.062	65.4	79.9	81.5
60	0.6	Skewed	-7%	-0.018	3.038	66.5	80.2	81.6
60	0.6	t_3	Null	-0.003	2.988	66.1	81.0	82.6
60	0.6	t_3	-3.5%	0.005	3.063	66.6	81.3	82.9
60	0.6	t_3	-7%	0.026	3.046	66.0	81.5	83.0

Table S23. Simulation results for $R = 120$ summers: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

R	ρ	Innovation	Effect	Bias	RMSE	Student	Lag 2	Growing
				percentage points		coverage (%)		
120	0.0	Gaussian	Null	0.013	1.098	94.8	94.3	93.7
120	0.0	Gaussian	-3.5%	0.003	1.100	94.9	94.4	93.7
120	0.0	Gaussian	-7%	0.023	1.101	94.7	94.2	93.2
120	0.0	Skewed	Null	0.011	1.091	94.4	93.7	93.0
120	0.0	Skewed	-3.5%	0.003	1.084	94.2	93.5	93.1
120	0.0	Skewed	-7%	-0.004	1.101	93.7	93.3	92.5
120	0.0	t_3	Null	0.008	1.115	95.2	94.5	93.9
120	0.0	t_3	-3.5%	-0.007	1.099	95.2	94.4	93.8
120	0.0	t_3	-7%	0.004	1.100	95.5	94.8	94.0
120	0.3	Gaussian	Null	0.013	1.493	84.8	90.8	91.5
120	0.3	Gaussian	-3.5%	0.001	1.471	85.6	91.2	91.8
120	0.3	Gaussian	-7%	0.002	1.489	84.8	90.9	91.5
120	0.3	Skewed	Null	-0.004	1.502	83.6	89.6	90.2
120	0.3	Skewed	-3.5%	-0.014	1.485	84.7	90.2	90.7
120	0.3	Skewed	-7%	0.025	1.491	84.2	90.0	90.4
120	0.3	t_3	Null	-0.009	1.495	84.7	91.1	91.7
120	0.3	t_3	-3.5%	-0.028	1.475	84.6	91.2	91.7
120	0.3	t_3	-7%	-0.002	1.488	84.5	90.7	91.4
120	0.6	Gaussian	Null	-0.018	2.164	67.9	82.8	86.5
120	0.6	Gaussian	-3.5%	-0.010	2.171	67.8	82.7	86.6
120	0.6	Gaussian	-7%	0.023	2.169	66.9	82.3	86.5
120	0.6	Skewed	Null	-0.012	2.162	67.2	81.8	85.6
120	0.6	Skewed	-3.5%	-0.020	2.181	65.7	81.3	85.6
120	0.6	Skewed	-7%	0.014	2.180	66.5	81.6	85.5
120	0.6	t_3	Null	-0.013	2.184	66.6	82.5	86.4
120	0.6	t_3	-3.5%	-0.053	2.146	66.5	82.5	86.6
120	0.6	t_3	-7%	0.015	2.180	65.7	82.2	86.3

Table S24. Simulation results for $R = 20$ summers: standardised-mean diagnostics. L is the growing Bartlett lag; the standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}, z_{.975}$
20	0.0	Gaussian	Null	2	0.006	0.990	-1.94, 1.98
20	0.0	Gaussian	-3.5%	2	-0.006	0.999	-1.98, 1.98
20	0.0	Gaussian	-7%	2	-0.006	0.993	-1.92, 1.99
20	0.0	Skewed	Null	2	-0.017	1.003	-1.71, 2.21
20	0.0	Skewed	-3.5%	2	0.008	0.997	-1.71, 2.18
20	0.0	Skewed	-7%	2	-0.009	0.992	-1.68, 2.14
20	0.0	t_3	Null	2	-0.010	0.996	-1.92, 1.95
20	0.0	t_3	-3.5%	2	0.011	0.968	-1.90, 1.90
20	0.0	t_3	-7%	2	0.010	1.005	-1.88, 1.91
20	0.3	Gaussian	Null	2	0.022	0.980	-1.91, 1.94
20	0.3	Gaussian	-3.5%	2	-0.026	0.983	-1.96, 1.93
20	0.3	Gaussian	-7%	2	0.010	0.990	-1.93, 1.92
20	0.3	Skewed	Null	2	0.014	0.988	-1.66, 2.18
20	0.3	Skewed	-3.5%	2	0.005	0.989	-1.68, 2.17
20	0.3	Skewed	-7%	2	-0.010	0.969	-1.68, 2.12
20	0.3	t_3	Null	2	0.015	0.953	-1.85, 1.91
20	0.3	t_3	-3.5%	2	-0.035	0.966	-1.94, 1.85
20	0.3	t_3	-7%	2	-0.001	0.990	-1.92, 1.91
20	0.6	Gaussian	Null	2	-0.008	0.950	-1.89, 1.83
20	0.6	Gaussian	-3.5%	2	0.000	0.956	-1.89, 1.85
20	0.6	Gaussian	-7%	2	0.000	0.964	-1.91, 1.88
20	0.6	Skewed	Null	2	0.007	0.967	-1.64, 2.10
20	0.6	Skewed	-3.5%	2	-0.008	0.958	-1.62, 2.11
20	0.6	Skewed	-7%	2	-0.007	0.953	-1.60, 2.11
20	0.6	t_3	Null	2	0.009	0.929	-1.77, 1.81
20	0.6	t_3	-3.5%	2	0.014	0.950	-1.81, 1.87
20	0.6	t_3	-7%	2	-0.005	0.979	-1.84, 1.79

Table S25. Simulation results for $R = 33$ summers: standardised-mean diagnostics. L is the growing Bartlett lag; the standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}, z_{.975}$
33	0.0	Gaussian	Null	3	0.027	0.988	-1.92, 1.97
33	0.0	Gaussian	-3.5%	3	-0.002	0.995	-1.95, 1.92
33	0.0	Gaussian	-7%	3	-0.010	0.987	-1.93, 1.93
33	0.0	Skewed	Null	3	-0.014	0.991	-1.75, 2.12
33	0.0	Skewed	-3.5%	3	-0.001	0.987	-1.72, 2.11
33	0.0	Skewed	-7%	3	0.002	0.996	-1.73, 2.13
33	0.0	t_3	Null	3	-0.009	1.000	-1.93, 1.92
33	0.0	t_3	-3.5%	3	0.003	0.987	-1.93, 1.93
33	0.0	t_3	-7%	3	-0.002	1.018	-1.94, 1.91
33	0.3	Gaussian	Null	3	-0.016	0.996	-1.99, 1.95
33	0.3	Gaussian	-3.5%	3	-0.001	0.990	-1.93, 1.94
33	0.3	Gaussian	-7%	3	0.006	0.987	-1.92, 1.92
33	0.3	Skewed	Null	3	0.000	0.982	-1.72, 2.11
33	0.3	Skewed	-3.5%	3	-0.001	0.995	-1.71, 2.13
33	0.3	Skewed	-7%	3	-0.003	0.981	-1.73, 2.09
33	0.3	t_3	Null	3	-0.004	0.974	-1.93, 1.89
33	0.3	t_3	-3.5%	3	0.004	0.985	-1.92, 1.94
33	0.3	t_3	-7%	3	0.012	0.970	-1.94, 1.91
33	0.6	Gaussian	Null	3	0.001	0.984	-1.92, 1.92
33	0.6	Gaussian	-3.5%	3	0.008	0.972	-1.89, 1.87
33	0.6	Gaussian	-7%	3	0.006	0.969	-1.90, 1.88
33	0.6	Skewed	Null	3	0.005	0.979	-1.70, 2.12
33	0.6	Skewed	-3.5%	3	0.010	0.970	-1.68, 2.10
33	0.6	Skewed	-7%	3	0.009	0.974	-1.68, 2.09
33	0.6	t_3	Null	3	-0.003	0.956	-1.87, 1.83
33	0.6	t_3	-3.5%	3	-0.013	0.958	-1.89, 1.82
33	0.6	t_3	-7%	3	0.001	0.995	-1.96, 1.88

Table S26. Simulation results for $R = 60$ summers: standardised-mean diagnostics. L is the growing Bartlett lag; the standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}$, $z_{.975}$
60	0.0	Gaussian	Null	3	0.001	1.003	-1.93, 1.95
60	0.0	Gaussian	-3.5%	3	-0.012	1.008	-2.01, 1.94
60	0.0	Gaussian	-7%	3	-0.002	1.001	-1.94, 1.98
60	0.0	Skewed	Null	3	-0.001	1.003	-1.79, 2.10
60	0.0	Skewed	-3.5%	3	0.013	0.995	-1.79, 2.09
60	0.0	Skewed	-7%	3	0.025	0.997	-1.78, 2.15
60	0.0	t_3	Null	3	0.013	1.022	-1.96, 2.01
60	0.0	t_3	-3.5%	3	0.005	0.983	-1.92, 1.96
60	0.0	t_3	-7%	3	-0.005	0.991	-1.89, 1.92
60	0.3	Gaussian	Null	3	0.008	0.993	-1.99, 1.92
60	0.3	Gaussian	-3.5%	3	-0.007	0.979	-1.92, 1.90
60	0.3	Gaussian	-7%	3	0.015	0.986	-1.91, 1.91
60	0.3	Skewed	Null	3	-0.009	0.988	-1.81, 2.07
60	0.3	Skewed	-3.5%	3	0.008	0.993	-1.78, 2.13
60	0.3	Skewed	-7%	3	-0.001	0.993	-1.83, 2.08
60	0.3	t_3	Null	3	-0.007	0.986	-1.92, 1.98
60	0.3	t_3	-3.5%	3	0.017	0.989	-1.91, 1.96
60	0.3	t_3	-7%	3	-0.005	0.988	-1.96, 1.95
60	0.6	Gaussian	Null	3	-0.003	0.986	-1.92, 1.94
60	0.6	Gaussian	-3.5%	3	-0.009	0.989	-1.96, 1.90
60	0.6	Gaussian	-7%	3	0.010	0.980	-1.89, 1.92
60	0.6	Skewed	Null	3	0.003	0.977	-1.78, 2.06
60	0.6	Skewed	-3.5%	3	-0.005	0.988	-1.77, 2.10
60	0.6	Skewed	-7%	3	-0.006	0.980	-1.76, 2.05
60	0.6	t_3	Null	3	-0.001	0.964	-1.90, 1.87
60	0.6	t_3	-3.5%	3	0.002	0.989	-1.89, 1.92
60	0.6	t_3	-7%	3	0.008	0.983	-1.94, 1.92

Table S27. Simulation results for $R = 120$ summers: standardised-mean diagnostics. L is the growing Bartlett lag; the standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	L	z mean	z SD	$z_{.025}$, $z_{.975}$
120	0.0	Gaussian	Null	4	0.012	1.003	-1.98, 1.97
120	0.0	Gaussian	-3.5%	4	0.002	1.004	-1.93, 1.99
120	0.0	Gaussian	-7%	4	0.021	1.005	-1.94, 2.02
120	0.0	Skewed	Null	4	0.010	0.996	-1.82, 2.07
120	0.0	Skewed	-3.5%	4	0.003	0.989	-1.86, 2.05
120	0.0	Skewed	-7%	4	-0.003	1.005	-1.88, 2.07
120	0.0	t_3	Null	4	0.007	1.018	-1.95, 2.01
120	0.0	t_3	-3.5%	4	-0.006	1.003	-1.96, 1.96
120	0.0	t_3	-7%	4	0.004	1.004	-1.92, 2.02
120	0.3	Gaussian	Null	4	0.008	1.000	-1.98, 1.95
120	0.3	Gaussian	-3.5%	4	0.001	0.986	-1.92, 1.94
120	0.3	Gaussian	-7%	4	0.001	0.997	-1.94, 1.95
120	0.3	Skewed	Null	4	-0.003	1.006	-1.86, 2.10
120	0.3	Skewed	-3.5%	4	-0.009	0.995	-1.87, 2.06
120	0.3	Skewed	-7%	4	0.017	0.999	-1.82, 2.06
120	0.3	t_3	Null	4	-0.006	1.001	-1.94, 1.93
120	0.3	t_3	-3.5%	4	-0.018	0.988	-1.95, 1.94
120	0.3	t_3	-7%	4	-0.001	0.997	-1.98, 1.94
120	0.6	Gaussian	Null	4	-0.008	0.988	-1.97, 1.95
120	0.6	Gaussian	-3.5%	4	-0.005	0.991	-1.97, 1.94
120	0.6	Gaussian	-7%	4	0.010	0.990	-1.92, 1.93
120	0.6	Skewed	Null	4	-0.005	0.987	-1.83, 2.09
120	0.6	Skewed	-3.5%	4	-0.009	0.996	-1.85, 2.06
120	0.6	Skewed	-7%	4	0.006	0.995	-1.82, 2.08
120	0.6	t_3	Null	4	-0.006	0.997	-1.87, 1.91
120	0.6	t_3	-3.5%	4	-0.024	0.979	-1.97, 1.90
120	0.6	t_3	-7%	4	0.007	0.995	-1.95, 1.93

Table S28. Simulation results for $R = 20$ summers: one-sided rejection percentages. For Student, lag-2 and growing-lag tests, null rows estimate mean-null size and non-null rows estimate power. The sign calculation targets $\Pr(Y_r < 0) = 1/2$; under centred skewed mean-null innovations, its rejection rate quantifies departure from the sign null. The three-test column requires Student, lag-2 and sign values all to be at most 0.025.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
20	0.0	Gaussian	Null	4.6	7.1	8.0	2.1	1.0
20	0.0	Gaussian	-3.5%	34.8	40.7	43.6	15.8	12.1
20	0.0	Gaussian	-7%	81.0	84.3	86.2	49.5	46.4
20	0.0	Skewed	Null	11.8	14.6	15.6	19.0	6.3
20	0.0	Skewed	-3.5%	45.4	49.5	51.6	53.8	31.2
20	0.0	Skewed	-7%	77.3	80.3	81.5	82.9	64.7
20	0.0	t_3	Null	5.2	7.5	8.4	2.2	0.9
20	0.0	t_3	-3.5%	43.9	49.8	52.3	32.7	21.2
20	0.0	t_3	-7%	85.7	87.7	88.8	82.8	71.0
20	0.3	Gaussian	Null	10.5	9.3	10.4	4.8	3.1
20	0.3	Gaussian	-3.5%	41.0	37.6	39.9	21.9	17.9
20	0.3	Gaussian	-7%	74.5	70.5	72.6	51.0	46.9
20	0.3	Skewed	Null	17.4	16.5	17.6	20.3	10.3
20	0.3	Skewed	-3.5%	47.8	44.6	46.3	50.9	33.3
20	0.3	Skewed	-7%	75.2	72.7	74.1	77.2	62.1
20	0.3	t_3	Null	11.3	10.0	11.0	5.5	3.1
20	0.3	t_3	-3.5%	48.8	44.7	46.9	35.2	25.9
20	0.3	t_3	-7%	81.3	78.0	79.7	75.4	64.6
20	0.6	Gaussian	Null	20.8	15.9	17.1	11.8	9.3
20	0.6	Gaussian	-3.5%	44.4	36.0	37.7	28.7	24.2
20	0.6	Gaussian	-7%	70.2	62.0	63.8	52.5	47.9
20	0.6	Skewed	Null	27.0	22.1	23.0	24.1	16.9
20	0.6	Skewed	-3.5%	51.7	44.9	46.5	47.2	36.5
20	0.6	Skewed	-7%	71.8	65.3	66.7	68.5	57.1
20	0.6	t_3	Null	21.5	16.0	17.2	12.6	9.4
20	0.6	t_3	-3.5%	49.2	40.8	42.4	36.9	29.6
20	0.6	t_3	-7%	76.2	68.8	70.4	66.9	58.6

Table S29. Simulation results for $R = 33$ summers: one-sided rejection percentages. For Student, lag-2 and growing-lag tests, null rows estimate mean-null size and non-null rows estimate power. The sign calculation targets $\Pr(Y_r < 0) = 1/2$; under centred skewed mean-null innovations, its rejection rate quantifies departure from the sign null. The three-test column requires Student, lag-2 and sign values all to be at most 0.025.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
33	0.0	Gaussian	Null	4.8	6.0	7.1	3.6	0.8
33	0.0	Gaussian	-3.5%	49.6	53.2	56.3	33.8	18.1
33	0.0	Gaussian	-7%	95.0	95.6	96.1	81.4	68.5
33	0.0	Skewed	Null	10.3	12.0	13.3	39.3	5.9
33	0.0	Skewed	-3.5%	55.5	58.1	60.6	84.7	42.7
33	0.0	Skewed	-7%	89.0	89.8	90.8	98.4	81.8
33	0.0	t_3	Null	5.0	6.5	7.8	4.0	0.7
33	0.0	t_3	-3.5%	58.2	61.0	63.6	62.0	33.2
33	0.0	t_3	-7%	94.4	94.9	95.6	98.3	87.9
33	0.3	Gaussian	Null	11.8	9.1	9.9	8.2	3.0
33	0.3	Gaussian	-3.5%	50.5	44.1	45.3	37.0	23.0
33	0.3	Gaussian	-7%	89.2	85.1	85.5	77.5	63.8
33	0.3	Skewed	Null	17.1	14.4	14.9	36.5	9.4
33	0.3	Skewed	-3.5%	55.7	50.7	51.6	76.8	40.7
33	0.3	Skewed	-7%	85.2	81.8	82.3	95.2	74.7
33	0.3	t_3	Null	11.7	9.3	10.0	9.1	3.3
33	0.3	t_3	-3.5%	57.8	51.2	52.2	57.5	33.7
33	0.3	t_3	-7%	90.7	87.9	88.2	94.3	79.5
33	0.6	Gaussian	Null	21.2	14.4	14.1	15.9	8.5
33	0.6	Gaussian	-3.5%	50.8	40.2	39.4	41.1	27.8
33	0.6	Gaussian	-7%	80.7	72.4	71.4	71.5	59.2
33	0.6	Skewed	Null	26.1	20.0	19.5	34.3	14.9
33	0.6	Skewed	-3.5%	55.5	46.7	45.9	65.2	39.0
33	0.6	Skewed	-7%	79.5	72.0	71.2	85.6	65.1
33	0.6	t_3	Null	21.6	14.4	14.1	16.5	8.1
33	0.6	t_3	-3.5%	56.9	46.7	45.8	52.6	34.8
33	0.6	t_3	-7%	84.8	77.6	76.8	84.6	68.4

Table S30. Simulation results for $R = 60$ summers: one-sided rejection percentages. For Student, lag-2 and growing-lag tests, null rows estimate mean-null size and non-null rows estimate power. The sign calculation targets $\Pr(Y_r < 0) = 1/2$; under centred skewed mean-null innovations, its rejection rate quantifies departure from the sign null. The three-test column requires Student, lag-2 and sign values all to be at most 0.025.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
60	0.0	Gaussian	Null	5.0	5.8	6.5	4.9	0.8
60	0.0	Gaussian	-3.5%	71.8	73.6	74.7	54.4	31.7
60	0.0	Gaussian	-7%	99.7	99.7	99.7	96.8	90.7
60	0.0	Skewed	Null	9.2	10.0	10.6	62.6	5.2
60	0.0	Skewed	-3.5%	70.6	71.7	72.7	98.3	59.5
60	0.0	Skewed	-7%	97.6	97.7	97.9	100.0	95.5
60	0.0	t_3	Null	4.7	5.5	6.2	4.7	0.5
60	0.0	t_3	-3.5%	77.5	78.2	79.1	87.0	57.0
60	0.0	t_3	-7%	98.8	98.7	98.8	100.0	97.6
60	0.3	Gaussian	Null	11.2	8.2	8.3	8.9	2.8
60	0.3	Gaussian	-3.5%	67.5	59.5	59.4	54.2	33.7
60	0.3	Gaussian	-7%	98.4	97.0	96.9	93.9	86.0
60	0.3	Skewed	Null	15.2	11.9	12.1	52.5	8.3
60	0.3	Skewed	-3.5%	67.5	61.3	61.1	93.7	52.2
60	0.3	Skewed	-7%	95.5	93.2	93.1	99.7	89.6
60	0.3	t_3	Null	11.8	8.1	8.1	10.0	2.4
60	0.3	t_3	-3.5%	71.9	64.7	64.5	77.5	47.9
60	0.3	t_3	-7%	97.4	96.5	96.3	99.6	93.9
60	0.6	Gaussian	Null	21.2	13.1	12.3	16.8	7.3
60	0.6	Gaussian	-3.5%	63.3	50.9	48.8	53.3	35.7
60	0.6	Gaussian	-7%	92.7	87.0	85.7	87.2	75.9
60	0.6	Skewed	Null	24.4	17.9	16.9	42.5	13.4
60	0.6	Skewed	-3.5%	64.4	54.2	52.3	80.7	46.7
60	0.6	Skewed	-7%	90.1	84.3	83.2	96.5	79.1
60	0.6	t_3	Null	21.6	13.6	12.5	17.7	7.4
60	0.6	t_3	-3.5%	66.9	56.5	54.2	66.6	43.3
60	0.6	t_3	-7%	92.9	88.4	87.5	95.5	83.1

Table S31. Simulation results for $R = 120$ summers: one-sided rejection percentages. For Student, lag-2 and growing-lag tests, null rows estimate mean-null size and non-null rows estimate power. The sign calculation targets $\Pr(Y_r < 0) = 1/2$; under centred skewed mean-null innovations, its rejection rate quantifies departure from the sign null. The three-test column requires Student, lag-2 and sign values all to be at most 0.025.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
120	0.0	Gaussian	Null	5.1	5.5	5.9	4.0	0.7
120	0.0	Gaussian	-3.5%	93.3	93.6	93.8	78.8	65.7
120	0.0	Gaussian	-7%	100.0	100.0	100.0	100.0	99.8
120	0.0	Skewed	Null	7.8	8.0	8.4	86.3	4.2
120	0.0	Skewed	-3.5%	90.3	90.5	90.8	100.0	83.8
120	0.0	Skewed	-7%	100.0	99.9	99.9	100.0	99.9
120	0.0	t_3	Null	4.9	5.2	5.6	4.2	0.6
120	0.0	t_3	-3.5%	93.5	93.6	93.9	98.8	87.1
120	0.0	t_3	-7%	99.7	99.8	99.8	100.0	99.6
120	0.3	Gaussian	Null	11.0	7.9	7.4	7.9	2.9
120	0.3	Gaussian	-3.5%	87.2	81.7	80.2	74.4	60.5
120	0.3	Gaussian	-7%	100.0	100.0	100.0	99.6	99.2
120	0.3	Skewed	Null	14.8	11.0	10.4	71.9	7.4
120	0.3	Skewed	-3.5%	84.6	79.2	77.9	99.7	71.6
120	0.3	Skewed	-7%	99.6	99.2	99.2	100.0	98.6
120	0.3	t_3	Null	12.0	7.9	7.4	8.8	2.3
120	0.3	t_3	-3.5%	88.8	84.5	83.4	94.2	75.5
120	0.3	t_3	-7%	99.7	99.4	99.4	100.0	99.1
120	0.6	Gaussian	Null	20.4	12.5	10.2	15.9	7.2
120	0.6	Gaussian	-3.5%	78.9	67.5	62.8	68.8	53.9
120	0.6	Gaussian	-7%	99.2	98.2	97.2	97.7	95.0
120	0.6	Skewed	Null	23.4	15.9	13.4	53.8	11.9
120	0.6	Skewed	-3.5%	77.5	67.9	63.2	94.3	60.0
120	0.6	Skewed	-7%	97.9	95.6	94.3	99.9	93.2
120	0.6	t_3	Null	21.9	13.5	10.9	17.3	7.2
120	0.6	t_3	-3.5%	81.0	71.7	67.3	83.3	62.1
120	0.6	t_3	-7%	98.7	97.4	96.7	99.8	96.0

An additional 10,000-replication experiment uses the exact 1991–2025 calendar and removes 2015 and 2022. The largest absolute mean bias over correlations 0, 0.3 and 0.6 and effects 0 and -7% is 0.03 percentage points. Excluding the two design summers therefore has negligible first-order centring bias, but short-record dependence remains: at correlation 0.6, null coverage is 68.3% for the Student interval that treats summers as independent and 80.2% for calendar lag 2.

S11.4. Dependence across monthly records

Proposition 1 uses joint invariance under independent rotations of all monthly records. The product condition additionally specifies their joint behaviour across records. The stress test uses the 121 observed site coordinates, the five Gaussian graphs and six records from two simulated summers. The June, July and August record lengths were 30, 31 and 31 days in each summer.

For summer y, let

$$U_y(u) = \sum_{k=1}^{3} \sqrt{w_k}\{A_{yk}\cos(2\pi ku) + B_{yk}\sin(2\pi ku)\}, \qquad u \in [0,1),$$

where the coefficients are independent standard normal variables and $(w_1, w_2, w_3) = (1, 0.45, 0.20)/1.65$. The same coefficients were used for all three records in a summer. For month m with length n_m, the centred spatial field at $t = 0, \ldots, n_m - 1$ was

$$\mathbf{z}_{ymt} = \lambda U_y(t/n_m)\mathbf{g} + \sqrt{1-\lambda^2}\,\boldsymbol{\varepsilon}_{ymt}, \qquad \lambda \in \{0, 0.3, 0.6, 0.8\}.$$

The vector g is the centred, unit-mean-square pattern proportional to $0.8x + y$ in projected coordinates. The month-specific Gaussian noise used the first 12 square-root-eigenvalue-weighted eigenmodes of the centred spatial covariance $0.95\exp(-d/450) + 0.05I$, where d is in km. The retained basis was rescaled to mean site variance one. Its circular moving-average filter had weights $(0.20, 0.50, 1, 0.50, 0.20)$, normalised to unit Euclidean norm. Regional-mean series X_{ymt} were generated from an independent random-number stream with a circular filter proportional to $(0.25, 0.60, 1, 0.60, 0.25)$. The simulated field is $\mathbf{y}_{ymt} = X_{ymt}\mathbf{1} + \mathbf{z}_{ymt}$; within-record type-7 quartiles of X_{ymt} define the high and middle labels.

Each individual record has a cyclically stationary Gaussian distribution. The shared Fourier coefficients align the three monthly phases within a summer, so independent month-wise rotations alter their cross-covariance when $\lambda > 0$. A numerical covariance check gave a maximum marginal one-step rotation error of 2.0×10^{-15}. Independently rotating the 30-day record by one day changed its cross-covariance with a 31-day record by 0.275 in relative Frobenius norm; the corresponding graph-dispersion cross-covariance changed by 0.362. The pooled same-summer July–August correlation of corresponding daily mean-dispersion values, averaged over the five graphs, increased from 0.003 at $\lambda = 0$ to 0.769 at $\lambda = 0.8$.

For each value of λ, we generated 2,000 replications and used 999 independent product shifts per replication. The statistic averaged the record-specific high-to-middle relative contrasts (ratios minus one) equally over the six records and five graphs. Table S32 reports lower-tail rejection at $\alpha = 0.05$; the seed was 20260812.

Table S32. Product-shift size under shared cross-record temporal dependence.

Shared loading λ	Rejection rate	Monte Carlo SE
0.0	0.0515	0.00494
0.3	0.0510	0.00492
0.6	0.0495	0.00485
0.8	0.0490	0.00483

Rejection remained between 4.90% and 5.15% as the shared component increased. Rejection rates remain near the nominal level across the four shared-process settings.

S12. Sensitivity to event thresholds

We compare six event partitions across the 99 evaluation month-year records. These exploratory analyses were added on 7 September 2026. For each record, replace the lower and upper quantile probabilities in the main-text event sets by (a, b). High days have $\bar{y}_t \geq q_{b,ym}$, middle days have $q_{a,ym} \leq \bar{y}_t < q_{b,ym}$ and low days are excluded. We retain the UTC peak fields, 121 sites, original distances, five bandwidths and equal-month, equal-scale and equal-summer aggregation. Linear sample quantiles are identical to type 7.

Table S33. Event-partition sensitivity over 33 evaluation summers. Entries are relative changes in percent; the interval is a descriptive 95% t-reference interval based on between-summer variation, calculated under independence of summers. V is ordinary spatial variance, evaluated with identical labels and equal-month/equal-summer aggregation. The continuous row reports within-record $\log Q_h$ slopes per °C, using all days.

(a, b)	Five-scale mean	Interval	126 km	2,013 km	V
$(.25, .75)$	-7.28	$[-10.00, -4.57]$	-2.86	-13.27	-14.45
$(1/3, 2/3)$	-6.48	$[-9.08, -3.89]$	-2.42	-12.05	-13.14
$(.20, .80)$	-8.48	$[-11.18, -5.78]$	-3.42	-15.04	-16.30
$(.30, .70)$	-6.88	$[-9.50, -4.26]$	-2.59	-12.65	-13.78
$(.25, .70)$	-6.90	$[-9.54, -4.26]$	-2.61	-12.78	-13.94
$(.25, .80)$	-8.27	$[-10.93, -5.61]$	-3.34	-14.52	-15.70
Continuous	-0.065	$[-0.082, -0.048]$	-0.032	-0.110	–

The five-scale mean is negative in 27–29 of the 33 summers across the six partitions. Tertile records contain 10 or 11 high days and 10 middle days. All six mean profiles become more negative at each successive bandwidth. Changing the cut points changes the contrast between the warm and reference states: the five-scale contraction ranges from 6.48% for tertiles to 8.48% for the 20th/80th-percentile partition. Ordinary variance also contracts under each partition. The direction and scale ordering persist across these threshold choices.

S13. Coverage estimation on the nested grid

S13.1. Spatial support of the coverage measures

This extension was specified on 7 September 2026 before calculating its coverage contrasts or comparison errors. It uses 3,036 synchronous daily fields in the 33 evaluation summers and retains the primary-grid peak time and original event labels. The dense fields contain 465 sites, including all 121 original sites and 344 additional locations. Embedded values agree with the primary fields to within 10^{-6} degrees Celsius. Every summer has 92 complete daily fields. The original graph contrast is reproduced with absolute numerical error below 10^{-10}.

Each site represents a cell extending 0.4° in longitude and 0.45° in latitude from its centre. Cells are clipped to $[105, 125]^\circ$E by $[20, 42]^\circ$N and the union of land polygons in the Natural Earth low-resolution map distributed with R package `spData`. All countries within the original rectangle are included. Retained cells represent 3,220,939 km^2, or 95.69% of the 3,366,047 km^2 of mapped land in the rectangle. The denominator uses the represented land. All 465 cells have positive land area. Area is computed on the WGS84 ellipsoid, including polygon holes. A coastal cell has one observed temperature and its clipped land area. Grid adjacency gives connectivity at this resolution; subcell coastline connectivity is unresolved.

Four-neighbour components use adjacent grid rows or columns; eight-neighbour components additionally join diagonal neighbours. Missing lattice positions remain absent. For each day and threshold, both measured curves satisfy $0 \le C_t(u) \le A_t(u) \le 1$ and decrease monotonically with u. The additional-site exceedance proportion uses the 344 new cells and their own normalised land areas. Connected area uses the complete retained lattice.

S13.2. Threshold dependence of coverage

The pooled 465-site values from 2015 and 2022 determine the temperature grid. Their 1st and 99.5th sample percentiles are rounded outward to multiples of 0.25°C, producing 9.5–28.5°C. All 77 thresholds use strict exceedance, $y_{it} > u$. These levels describe the temperature field. For $F = A$ or C, the contrast is

$$\Delta_F(u) = \frac{1}{33} \sum_{y \in \mathcal{Y}_H} \frac{1}{3} \sum_{m \in \{6,7,8\}} \left\{ \frac{\sum_{t \in \mathcal{E}_{ym}} F_t(u)}{|\mathcal{E}_{ym}|} - \frac{\sum_{t \in \mathcal{M}_{ym}} F_t(u)}{|\mathcal{M}_{ym}|} \right\}.$$

All annual curves and both group means are retained. Square-kilometre contrasts multiply these differences by the represented land area.

Descriptive uncertainty uses 1,999 circular moving-block resamples of the 35 calendar positions from 1991 to 2025, with block length five and seed 20260907. Seven uniformly drawn block starts produce 35 positions per resample. Development-year positions are then omitted and the remaining annual contrasts averaged. All thresholds and outcomes share the same draws. The 2.5th and 97.5th percentiles form pointwise intervals. Paired model-error intervals use the same resampling and condition on the fitted cross-validation predictions. They describe variation in the observed sequence; they are pointwise rather than simultaneous bands.

Eight-neighbour adjacency changes the high-minus-middle connected-area contrast by at most 0.965 percentage points over the threshold grid. Replacing clipped land areas by normalised cosine-latitude weights changes the total-area and connected-area contrasts by at most 0.484 and 0.611 points, respectively. The two alternating 0.5°C subgrids retain the overall model ordering for all-day errors.

S13.3. Coverage at the illustrative threshold

High-regional-mean days have greater simultaneous exceedance area and larger connected exceedance regions throughout the evaluated temperature range (main-text Figure 4). At 24°C, the mean exceedance proportion rises from 37.39% on middle days to 50.29% on high days. The 12.91-percentage-point difference corresponds to approximately 416,000 km^2 (pointwise 95% block-resampling interval: 377,000–459,000 km^2). The largest connected region rises from 32.10% to 44.67% of represented land, a difference of 12.57 points or approximately 405,000 km^2 (365,000–452,000 km^2). Both differences are positive in all 33 summers at this threshold.

S13.4. Regression specification

The baseline inputs are the original regional mean and day of summer. The strong simple model adds V_t, the signed ordinary least-squares slope of WBT against latitude, and the 121-site direct exceedance proportion. Each dense cell's area is assigned to its nearest original site in the original projected kilometre coordinates to obtain fixed primary representative weights. Thus the direct proportion covers the same represented support using 121 temperatures.

The graph model adds $\log\{Q_{h_k}(\mathbf{y}_t)/V_t\}$ for all five original physical bandwidths. The variogram model adds the corresponding log ratios of five binned semivariances to V_t. Bins use the quintiles of the 7,260 original pair distances: internal boundaries are 600.71, 850.88, 1,201.42 and 1,553.64 km. Ties stay in the same bin; bin sizes are 1,422, 1,471, 1,454, 1,458 and 1,455. Every pair belongs to one bin. All observed variances and energies are positive. The specified constant-field convention sets the five log-shape inputs to zero when $V_t = 0$.

Each baseline scalar has a quadratic B-spline basis with four equally spaced knots over its training range, linear extrapolation and one redundant basis column removed. A constant training variable is omitted from the spline basis. Graph and variogram shapes enter linearly. All features are standardised using training observations. Weighted ridge least squares fits an unpenalised intercept and penalised slopes; training weights give equal weight to months within years. Predicted proportions are clipped to $[0, 1]$. The same procedure is used for total area, connected area and the additional-site area. One penalty per method, endpoint and outer fold is selected from $\{0.1, 1, 10, 100, 1000\}$ by inner mean absolute error averaged equally across thresholds, months and years.

The interpolation comparison uses a Delaunay piecewise-linear interpolant in the original projected coordinates. Eight dense locations lie outside the convex hull and use their nearest original site's temperature. Coverage is then calculated directly from the interpolated field. Interpolation preserves the original-site values to numerical precision.

The regressions estimate each threshold and outcome separately. Across all held-out day–threshold pairs, fitted connected area exceeds fitted total area in 0.31%, 1.44%, 1.59% and 1.78% of predictions for the mean, simple, variogram and graph models. The maximum graph-model excess is 3.30 percentage points. Adjacent-threshold increases occur in 2.23% of graph area predictions and 7.49% of graph connected-area predictions. These discrepancies describe approximation error in the regression summaries; measured and interpolated fields satisfy the coverage identities.

S13.5. Temporal validation

The outer blocks are 1991–1995, 1996–2000, 2001–2005, 2006–2010, 2011–2015, 2016–2020 and 2021–2025. Each is held out in turn. Training excludes the held-out block, its adjacent calendar years and the two development summers. Inner validation uses the remaining calendar blocks within outer training, again excluding adjacent years. Spline knots, standardisation and penalties are fitted within these training sets. All methods share the same memberships. Training may use years before and after the held-out block; this evaluates transport across omitted years.

For method j, summer y and outcome F, define

$$L^F_{j,y} = \frac{1}{3} \sum_{m\in\{6,7,8\}} \frac{1}{|\mathcal{T}_{ym}|} \sum_{t\in\mathcal{T}_{ym}} \frac{1}{77} \sum_{u\in\mathcal{U}} |F_t(u) - \widehat{F}^{(-b(y))}_{j,t}(u)|.$$

High-day errors replace $\mathcal{T}_{ym}$ by $\mathcal{E}_{ym}$. Paired differences subtract graph error from the strong-simple or variogram error within each summer. Every method supplies 3,036 held-out daily curves. Positive differences indicate lower graph error.

Table S34. Paired reduction in held-out mean absolute error after adding graph information, in percentage points. Positive values favour the graph model. Intervals use five-year calendar-block resampling of the annual paired errors, conditional on the fitted cross-validation predictions.

Outcome	Days	Reference	Reduction	95% interval	Improved summers
Area	all	simple	0.0069	$[0.0057, 0.0081]$	28/33
Area	all	variogram	-0.0053	$[-0.0070, -0.0038]$	4/33
Area	high	simple	0.0041	$[0.0005, 0.0082]$	21/33
Area	high	variogram	-0.0017	$[-0.0046, 0.0010]$	12/33
Additional sites	all	simple	0.0077	$[0.0062, 0.0090]$	28/33
Additional sites	all	variogram	-0.0070	$[-0.0087, -0.0054]$	4/33
Additional sites	high	simple	0.0043	$[0.0006, 0.0081]$	20/33
Additional sites	high	variogram	-0.0029	$[-0.0062, 0.0005]$	12/33
Connected area	all	simple	0.0100	$[0.0069, 0.0129]$	26/33
Connected area	all	variogram	-0.0137	$[-0.0172, -0.0101]$	2/33
Connected area	high	simple	0.0036	$[-0.0033, 0.0100]$	18/33
Connected area	high	variogram	0.0016	$[-0.0028, 0.0060]$	18/33

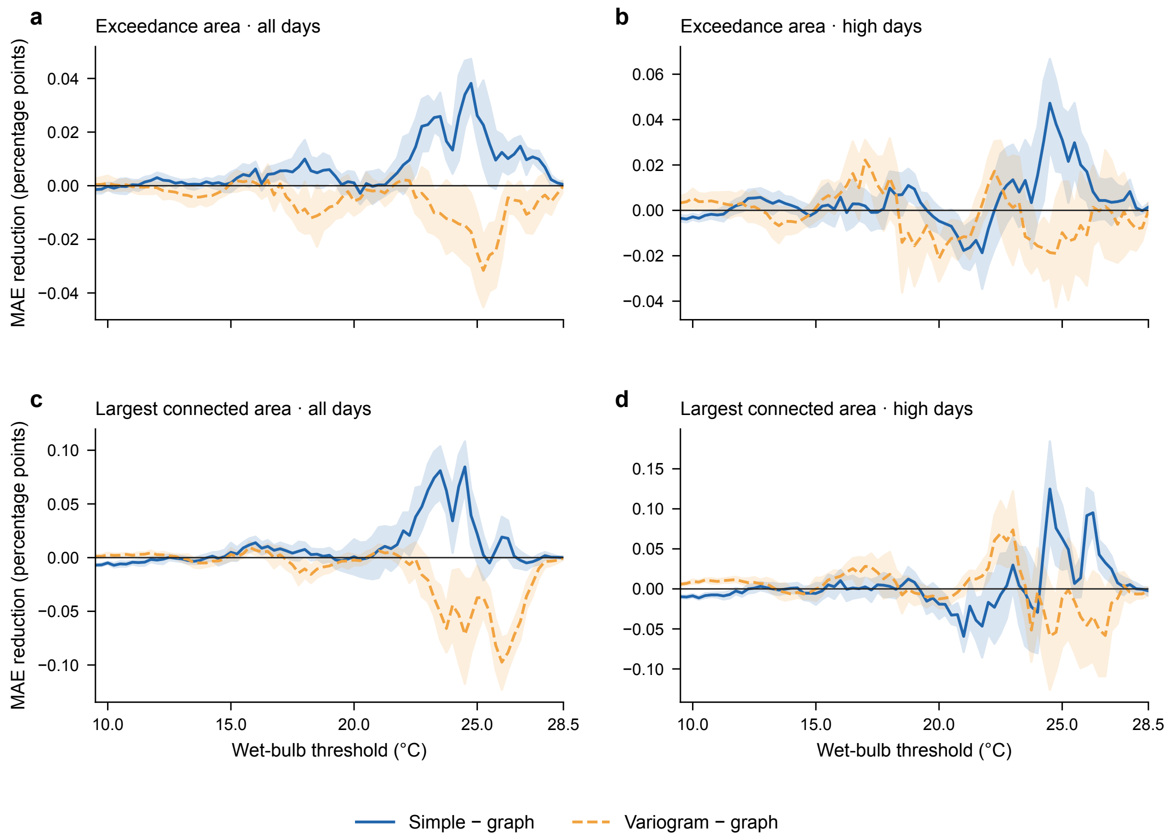

Figure S3 Threshold-specific paired reduction in mean absolute error from adding graph information. Positive values favour the graph model. Blue solid curves compare the strong simple baseline with the graph model; orange dashed curves compare the variogram model with the graph model. Left panels use all days and right panels use high days. Shading gives pointwise 95% five-year calendar-block resampling intervals, conditional on fitted predictions. All 77 thresholds are retained.

S14. Data preparation

S14.1. Thermodynamic conversion

WBT is calculated from temperature T and dew point T_d in kelvin and surface pressure p in pascals. Vapour pressure is $e(T_d) = 611.2\exp\{17.67(T_d - 273.15)/(T_d - 29.65)\}$ Pa, and the Bolton mixing-ratio, lifting-condensation-temperature and equivalent-potential- temperature relations are evaluated in these units (Bolton, 1980). Saturated T_w is obtained by solving

$$\theta_e(T_w, T_w, p) = \theta_e(T, T_d, p) \tag{S10}$$

at the original pressure with 40 bisection iterations on $[T_d, T]$; even a 100-K initial interval then falls below 10^{-9} K. Values with $T_d > T$ are clipped at equality before evaluation. The Stull approximation provides a comparison (Stull, 2011); the main calculation follows the pseudoadiabatic definition discussed by Davies-Jones (2008).

S14.2. Quality control

Every 16th longitude and 18th latitude cell from a fixed north-west origin gives a 13×13 candidate lattice; removing ocean cells leaves 121 sites (main-text Figure 2). The 35 summers contain 9,350,880 quality-controlled hour–site records and 92 complete UTC fields per summer. Because of NetCDF packing, 2,092 dew-point values exceeded air temperature slightly (0.022% of records); these pairs were clipped at equality before WBT was calculated.

S14.3. Alternative daily fields

For each UTC day, the 121 simultaneous values at the peak regional mean form the daily field. Within each month-year, the upper regional-mean quartile is high, the middle half is middle and the lower quartile is excluded. Labels depend on regional mean WBT, and the selected hour maximises that mean; graph dispersion enters after selection. UTC+8 days, alternative thresholds, sitewise maxima, all 24 fixed UTC hours and daily-mean fields provide comparisons. At each fixed hour, both the original peak-based labels and labels recomputed from that hour's regional mean are analysed; daily-mean labels are likewise recomputed from daily-mean WBT. A dated event manifest records the type-7 thresholds, membership and absence of ties.

Three checks address within-month seasonal progression: comparison of the day-of-month distributions, removal of a separate linear day-of-month trend at every site within each year–month, and subtraction of each site's calendar-day climatology, estimated from all summers except the one being analysed. The last two calculations use the original event labels. A date-matched calculation compares each high day with all middle days in the same month-year lying within ± 3 or ± 5 calendar days. It first averages the available high-day-specific ratios and then uses the same equal-month, equal-scale and equal-summer hierarchy as the estimator; unmatched-day and empty-record counts are retained.

A continuous diagnostic regresses $\log Q_h$ on centred regional mean WBT within each month-year and bandwidth. Slopes are then averaged equally over months, bandwidths and evaluation summers. This slope supplements the quartile target with a graded summary.

S14.4. Spatial weighting

The main target is a site average. For area weighting, $a_i = \cos(\mathrm{latitude}_i)$,

$$\bar{y}_t^{(a)} = \frac{\sum_i a_i y_{it}}{\sum_i a_i}, \qquad Q_{h,a}(\mathbf{y}) = \frac{\sum_{i<j} a_i a_j w_{ij,h} (y_i - y_j)^2}{2 \sum_{i<j} a_i a_j w_{ij,h}}.$$

The first area-weighted calculation replaces Q_h by $Q_{h,a}$ using the original labels; the second redefines the labels from $\bar{y}_t^{(a)}$ at the regional peak hours. The two calculations separate spatial weighting from event relabelling.

Alternative graph kernels are $\exp(-d_{ij}/\rho)$ and $[1 - d_{ij}/\rho]_+^2$. For each Gaussian scale, ρ is found by bisection so that the alternative kernel has the same edge-weighted mean pair distance. Across the five scales, their effective edge counts range from 334 to 7,112 and from 382 to 7,113, compared with 366 to 7,109 for the Gaussian graphs; the full matched-distance and effective-edge metadata accompany the analysis.

References

Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59, 817–858.

Bolton, D. (1980). The computation of equivalent potential temperature. *Monthly Weather Review*, 108, 1046–1053.

Davies-Jones, R. (2008). An efficient and accurate method for computing the wet-bulb temperature along pseudoadiabats. *Monthly Weather Review*, 136, 2764–2785.

Ibragimov, I.A. and Linnik, Y.V. (1971). *Independent and Stationary Sequences of Random Variables*. Groningen: Wolters-Noordhoff.

Newey, W.K. and West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55, 703–708.

Stull, R. (2011). Wet-bulb temperature from relative humidity and air temperature. *Journal of Applied Meteorology and Climatology*, 50, 2267–2269.